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LInking accounts for ecosystem Services and Benefits to the Economy Through bridging (LISBETH)
Part III

Integrating Ecosystem Services into Economic and Financial Decision-Making with a focus on Sustainable Development Goals

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Contents

Abstract 2

Acknowledgements 3

1 Introduction 4

2 An accounting definition of Natural Capital based on Ecosystem Services 7

2.1 What is an ecosystem service (flow/ transaction) 8

2.2 What is an asset (stock) 12

2.3 Summary 13

3 Different ways to use ES accounts and how they relate to the Sustainable Development Goals 15

3.1 Definition and importance of the Sustainable Development Goals 15

3.2 Use the ES accounts for conventional descriptive analysis 17

3.3 Use of ES accounts in combination with other data to produce novel indicators 18

3.4 Use ES accounts as an input variable in existing tools/ models/ frameworks 20

3.5 Summary 21

4 Ecosystem Services accounts in descriptive analysis 23

4.1 Descriptive analysis of ES in Europe 24

4.2 The GBF Headline indicator B.1: Ecosystem Services 25

5 Ecosystem Services accounts used to process new indicators 29

5.1 Dependence of fish supply on seagrass 29

5.2 Ecosystem Services Vulnerability Index for the agricultural sector 32

6 Ecosystem Services accounts as input data into economic models and tools 35

6.1 Dependency of firms on ecosystems services 35

6.2 Ecosystem services in a simple macroeconomic framework 38

6.3 Gross Ecosystem Product in a general equilibrium model 41

6.4 Natural capital and regional growth 46

7 Conclusions: paving the way forward 50

References 52

List of abbreviations and definitions 56

List of figures 57

Abstract

The acronym LISBETH stands for LInking ecosystem Services and Benefits to the Economy Through bridging. Its purpose is to explore the applications of ecosystem services (ES) accounts, which measure the contribution of ecosystems to human well-being and economic growth. This is the third of a series of technical reports that provide concrete examples of applications, including descriptive analysis, processing of new indicators, and using ES accounts as input variables in economic models. LISBETH part III presents some applications of ES accounts, including measuring the contribution of ecosystems to human activities and identifying their dependence on ES in line with the Sustainable Development Goals (SDGs). It also discusses the use of ES accounts in combination with other data to process new indicators, such as the dependence of fish supply on seagrass and the vulnerability of the agricultural sector to ES. In addition, the report explores the use of ES accounts as input variables in economic models, including the calculation of the Gross Ecosystem Product, the integration of ES into macroeconomic frameworks, and the potential of ES accounts to support the development of sustainable and inclusive economic growth strategies. Overall, the report aims to promote the use of ES accounts and facilitate a measurement system linked to the SDGs. By providing a comprehensive overview of the applications of ES accounts, the report aims to support the development of sustainable and inclusive economic growth strategies that take natural capital into account.

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1 Introduction

The sustainable development is a paramount global goal, with the United Nations' Sustainable Development Goals (SDGs) serving as a guiding framework for achieving a more equitable and environmentally conscious future (UN, 2015). In the European Union, the European Green Deal has set the stage for a transformative shift towards a low-carbon, resource-efficient economy. Ambitious policies are in place but the implementation of their targets still faces many challenges ahead (Marelli et al., 2025). The new European Union's priorities envisage a prosperous and competitive Europe, making a success of the green and digital transitions, as stated in the EU Strategic Agenda 2024-2029[1].

At the core of the plan for Europe's prosperity is the pivotal concept of "competitiveness", as outlined in the Draghi report (European Commission, 2024) and in the recent communication "A Competitiveness Compass for EU"[2]. This concept is driven by three key imperatives: closing the innovation gap, establishing a roadmap for decarbonisation based on clean technology, and enhancing security and reducing dependencies. The policy guidelines underpinning competitiveness take into account the decarbonisation of the economy, together with a shift towards a more sustainable pattern of production and consumption, with the aim of preserving the value of resources in the economy in the long term, considering climate regulation and ensuring food and water security.

Ultimately, every human activity depends directly or indirectly on nature, and therefore competitiveness must be sustainable to remain stable in the long term. The major challenge remains to identify and measure the critical linkages between nature and human activities (Dasgupta, 2021).

Building on this premises, this report explores the opportunities and challenges associated with integrating ecosystem services into economic and financial decision-making, with a focus on supporting the achievement of the SDGs in the EU.

Among the tools created to operate and measure such connection, there is the System of integrated Environmental and Economic Accounts (SEEA) lead and coordinated by the United Nations Statistical Division, whose purpose is to coherently and consistently connect a series of nature-related information to economic accounts (United Nations, 2024). One specific set of information concerns Ecosystem Services (ES) accounting, which measure in terms of yearly flow the contribution of ecosystems to human activities.

To produce accounts of ecosystem services (ES), practitioners need consistent approaches, methodologies, models and data. Given the complexity of the task, all steps from choosing the conceptual approach and the methodologies that most consistently reflect it, to finding available data, are critical. A substantial body of literature exists that addresses the concepts underlying ES (Costanza et al., 2017; de Groot et al., 2002; Gómez-Baggethun et al., 2010), the possible methodologies and models for assessing ES (Burkhard & Maes, 2017), and for accounting for ES (Bagstad et al., 2021). Good practice would necessitate the integration of these elements in a

[1] Ref. https://www.consilium.europa.eu/media/yxrc05pz/sn02167en24_web.pdf

[2] Ref. https://ec.europa.eu/commission/presscorner/detail/en/ip_25_339

consistent manner, thereby facilitating the comprehension of the potential applications of these accounts by those who require their use.

It is important to note that the compilation of ES accounts is one process, and the use of ES accounts is another. The purpose of this report is to explore how the use of ES accounts can take place once they are completed. In comparison to previous LISBETH (Linking ecosystem Services and Benefits to the Economy Through bridging) reports, the present research on uses and applications does not constitute an exploratory exercise on the possible integration with some economic tools (La Notte et al., 2020) or on the needs of the financial sector for environmental metrics (La Notte et al., 2022). In this report practitioners, who are not familiar with ES accounts, find concrete cases. These practitioners are familiar with economic and financial tools and wish to explore the feasibility of integrating ES accounts to address concrete issues raised by policy actors. Therefore, it is possible to identify three subjects: ES accounts compilers, users and the policy actors. Two elements are evident:

- ES accounts compilers need to be familiar with the approaches, methodologies and techniques needed to assess, to value and to account for ES; ES users do not but they need to be familiar with the approaches, methodologies and techniques that characterize their policy domain;
- ES accounts users are not necessarily the same as the policy actors. For the policy makers and companies to be able to make effective use of the accounts, a technical intermediate data processing step may be necessary.

In this report, concrete applications will illustrate all the different cases mentioned above. Specifically, as schematically summarised in Figure 1, the basic measurements that characterize the calculation of ES, which are assessed as appropriate, into an accounting mechanism framework. The ES accounts can then be provided:

- directly to users, as in the case of statistical offices, international organisations and departments that directly process indicators (for example at regional level);
- to other compilers who will use the ES accounts to produce the results that are ultimately needed by the final policy actors, as in the case of the Ministries of Economy and Finance, the Central Banks, the Departments of Land Management, Agriculture, Forestry and Fisheries.

Figure 1. Stylized visualization on how ES accounts can directly support some policy actors, such as statistical offices, policy actors that directly process indicators and international organizations that collect data; or indirectly support other policy actors through the work of other experts working in specific disciplinary areas such as macroeconomics, central banks, agricultural and fisheries policies

Source: own elaboration

In the case of both direct and indirect support, the exchange between ES account providers and ES account users' needs to be participatory, at least in this initial phase when a use routine is not yet established. Indeed, on the one hand, the interpretation of some ES flows and the meaning of their trend over time may not be straightforward and intuitive. On the other hand, those compiling ES accounts may not have a full understanding or mastery of the economic and financial tools that the use of ecologically extended accounting requires.

After the conceptual explanation of the accounting definition of natural capital based on an ES framework (chapter 2) and the three possible work streams of ES accounts (chapter 3), concrete applications using the INCA ES accounts are presented (chapters 4 to 6). They constitute a starting point that can provide insights into the way ahead for a more inter- and trans-disciplinary research agenda that embeds both micro and macroeconomic perspectives at different territorial and administrative scales (chapter 7). Another application focuses on the Sustainable Development Goals, and the three dimensions of financial sustainability: environmental, social and economic.

2 An accounting definition of Natural Capital based on Ecosystem Services

On March 2021 the United Nations Statistical Commission adopted the System of integrated Environmental and Economic Accounting – Ecosystem Accounting (SEEA EA)[3] . The SEEA EA component complements the System of integrated Environmental and Economic Accounting – Central Framework (SEEA CF) the standard for which was adopted in 2012[4]. Both the SEEA EA and CF are satellite accounts in relation to the System of National Accounts (SNA) with which they share the accounting rules and mechanism.

In Europe, the experimental version of the SEEA EA (United Nations, 2019; United Nations et al., 2014) was tested and implemented through the INCA (Integrated system for Natural Capital Accounts) project, a partnership between the Joint Research Centre, the European Environment Agency, DG Environment, DG Research and Innovation, coordinated by Eurostat. The lessons learnt through INCA (La Notte et al., 2022) proved to be very useful at the international level by providing insights (Vysna et al., 2021) into the SEEA EA revision that eventually led to the SEEA EA standards (United Nations, 2024), and at the European level by providing a solid bases for the current EU legislation (Regulation 601/2011) that introduces ecosystem accounting as a mandatory requirement (Annex IX in (European Parliament and Council, 2024)).

From an accounting point of view, each item must fit with the others and all must be consistent within the same framework. Such coherence and internal consistency ensure that those who use SNA for their analyses can also extend their data sources to the SEEA components.

However, given the complexity of ES, coherence should include a careful understanding of the different ways in which SEEA CF deals with individual natural resources and SEEA EA deals with the whole ecosystem and its services. SEEA CF and SEEA EA are not mutually exclusive but rather complementary. However, this complementarity is often not obvious and needs to be presented in a clear and explicit way.

There are in fact two key concepts that need clarification:

1. the definition of services as positive transactions provided by ecosystems to socio-economic systems, and how to deal with flows that are not positive transactions:
    - the lack of ES (transactions);
    - the negative externalities generated by economic activities;
    - the disservices.
2. the definition of assets with respect to:
    - natural resources;
    - ecosystems.

[3] Ref. 52/108 in https://unstats.un.org/unsd/statcom/52nd-session/documents/2021-30-FinalReport-E.pdf

[4] Ref 43/105 in https://unstats.un.org/UNSDWebsite/statcom/session_43/documents/statcom-2012-43rd-report-E.pdf

While the present chapter remains theoretical in nature in terms of explaining the concepts, each of the cases mentioned herein will be recalled in the concrete applications reported in the following chapters as appropriate.

2.1 What is an ecosystem service (flow/ transaction)

In an accounting framework, ES are defined as the contribution of ecosystems to human activities, which include economic sectors, governments, households and the rest of the world. Beyond the accounting framework, the contribution of ES would go beyond human activities to include human well-being, which is a much broader concept.

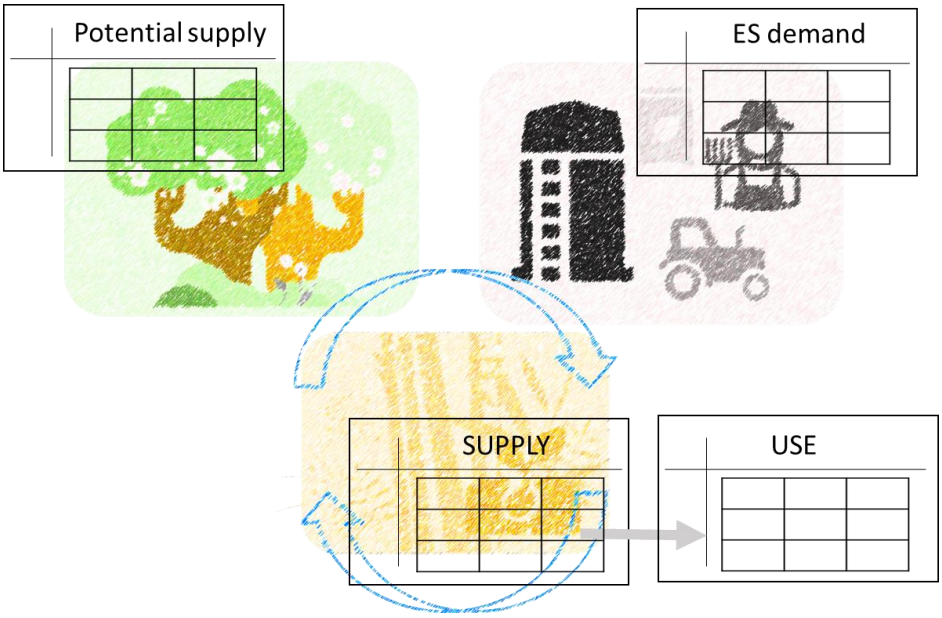
ES are generally classified as provisioning, regulating and maintenance, and cultural (Haines-Young & Potschin, 2018). Examples of provisioning services are timber and fish provision; examples of regulating and maintenance services are pollination, soil retention, flood control, carbon sequestration, air filtration, water purification, habitat and species maintenance; examples of cultural services are nature-based recreation and education, mimicking nature.

The concept of ecosystem contribution embeds all possible ecological processes that operate within the ecosystem to make it function. When the contribution is directed to humans, it becomes a "service" because it serves a human purpose; more specifically, the human activities that characterise the accounting framework can make the purpose very specific within a given production-consumption-accumulation boundary.

Therefore, two interacting components create a "service": the ecosystem component that has the potential to provide support, and the socio-economic component that has the need for that support (La Notte et al., 2019). The service that is generated is delivered as a positive transaction to economic sectors, governments, households and the rest of the world. Conceptually, there are two ES *supplies* that should not be mixed, confused or overlapped (Figure 2):

- The potential supply that ecosystems could provide, but that is not yet delivered
- The actual supply that results from the interaction with the demand expressed by the socio-economic system, which is delivered.

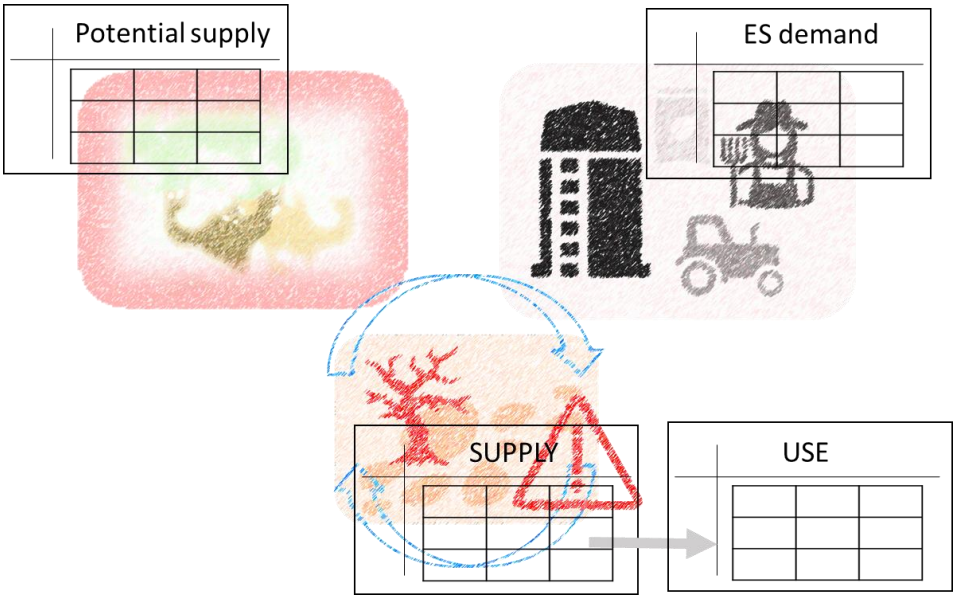
Figure 2. Ecosystem Service (ES) actual flow generated by the interaction between the potential supplied of the ecosystem and the demand from the socio-economic system.



Source: own elaboration

If the ES potential is higher than the ES demand, this is not a problem, but rather an advantage, because the carrying capacity of the ecosystem is enough to provide services over time. If the ES potential is lower than the ES demand, this is a problem because there is a human need that is not being met. The application of ES accounts within the INCA project carried out by the JRC has explicitly addressed this issue and has collected specific lessons learned (La Notte, et al., 2022) on how to report the mismatch between ES potential and ES demand, depending on the causes that may originate the mismatch (i.e. overuse, unmet demand and missed flows). Mismatches (with ES potential < ES demand) imply negative flows, and when they occur, they should be reported, but not in the accounting table that reports the positive transactions from ecosystems to socio-economic systems. A complementary table can be built to complete the information and provide the basis for monitoring over time not only the actual ES flows delivered, but also the ES vulnerabilities left open by the mismatches (Figure 3).

Figure 3. Ecosystem Service (ES) vulnerability generated by the mismatch between the ES potential and the ES demand, when the ES demand is higher than the ES potential. A complementary Supply and Use table can be compiled



Source: own elaboration

A different perspective applies to the concept of negative externalities. In economic theory, negative externalities are defined as a type of market failure that occurs when production or consumption activities have a negative impact on third parties, such as the emission of pollutants into the environment. Like all negative externalities, pollution is an unintended consequence of production and consumption activities and is not reflected in the market price of production and consumption activities.

The relevant data for this type of negative externality are emissions to air, soil and water and the generation of waste. In the SEEA CF there are specific accounting modules for emissions and waste.

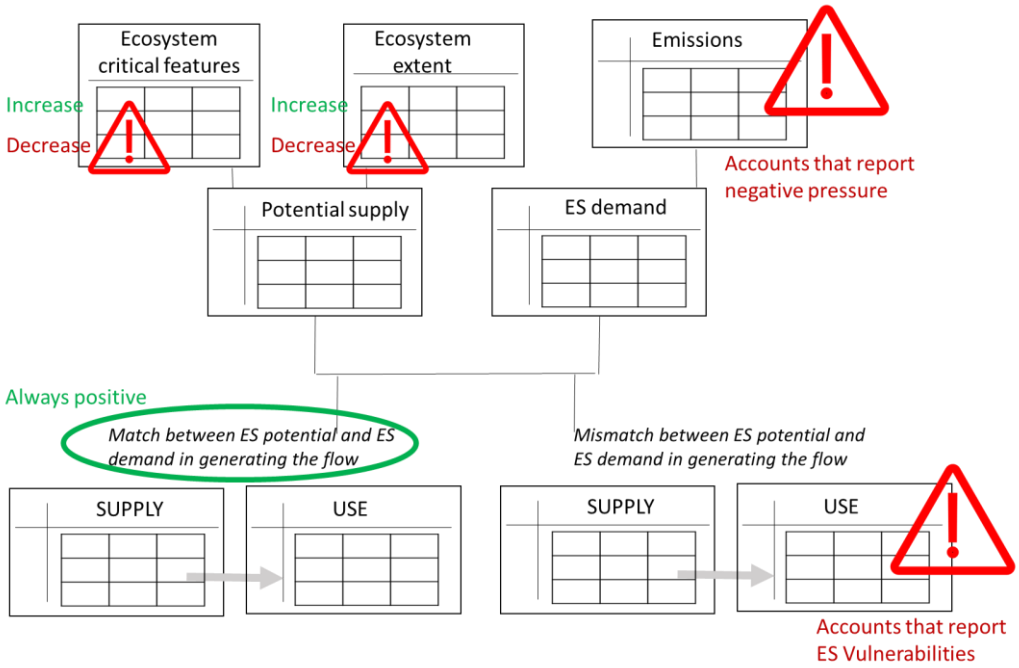
From an ES perspective, on the one hand there are the sink services: air filtration, water purification and soil decontamination which specifically measure the role of ecosystems in absorbing/cleaning/removing pollutants. Global climate regulation through carbon storage and sequestration also contributes to climate change mitigation. On the other hand, waste and pollution degrade ecosystems and ultimately their capacity to provide services.

The way in which negative externalities can be taken into account is made:

- indirectly through the emission and wastes accounts described in SEEA CF and whenever possible used as input variable in the biophysical modelling of sink ES;
- directly through the assessment and accounting of sink ES;
- indirectly through the pollution and waste-driven degradation of ecosystems that are not able to provide the same quantity and quality of ES over time.

As further highlighted in Figure 4, negative externalities are reported in different ways in different tables, but not in the accounting table that reports the positive transactions from ecosystems to socio-economic systems.

Figure 4. Tables that form the overall accounting framework: red signs indicate where to account for factors that may negatively affect the production of ecosystem services and determine a shortfall in provision.



Source: own elaboration

It should be noted that sink ES flow measurement can provide misleading results when considered in isolation. In fact, the service flow increases with the amount of pollutants. Therefore, increasing sink ES flow is not a policy message that supports sustainability. As water purification accounting has shown in the past (La Notte et al., 2017), the actual flow of ES tells one side of the story, i.e. how much inland water ecosystems support nitrogen removal in Europe. However, to complete the story, it is necessary to measure how much of this ES flow is sustainable over time. The difference between the actual ES flow and the sustainable ES flow will provide the ES overuse metric described above.

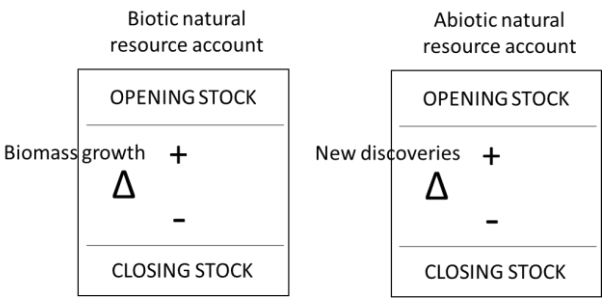
Finally, the notion of disservice would be an additional (in our context unnecessary) flow. For example, some ecosystems, such as wetlands or algal blooms, can produce strong, unpleasant odours. Similarly, certain birds or insects can produce noise that interferes with human activities. In addition, some plants produce pollen that can aggravate respiratory problems such as allergies and asthma. These are ecological processes that could be harmful rather than beneficial. They are not currently included in any ES classification and are often confused with a lack of ES or with poor management of ecosystems and negative externalities. This report does not consider disservices.

2.2 What is an asset (stock)

The definition of assets takes on different nuances when applied to the natural resources recorded in the SEEA CF and to the ecosystems recorded in the SEEA EA. Although the definitions are complementary, it is important to emphasise that they are different.

The SEEA CF provides accounting formats for both biotic and abiotic resources. As shown in Figure 5, the asset account is structured according to an opening stock referring to the beginning of the accounting period (usually one year) which is subject to additions and reductions (during the accounting period) the aggregation of which results in the closing stock (at the end of the accounting period). In the case of biotic resources (e.g. timber and fisheries) the additions are represented by the natural growth in biomass, which have a direct correspondence with provisioning ES. In the case of abiotic resources (e.g. subsoil and subsea assets) the additions are represented by new discoveries, which do not have a corresponding ES.

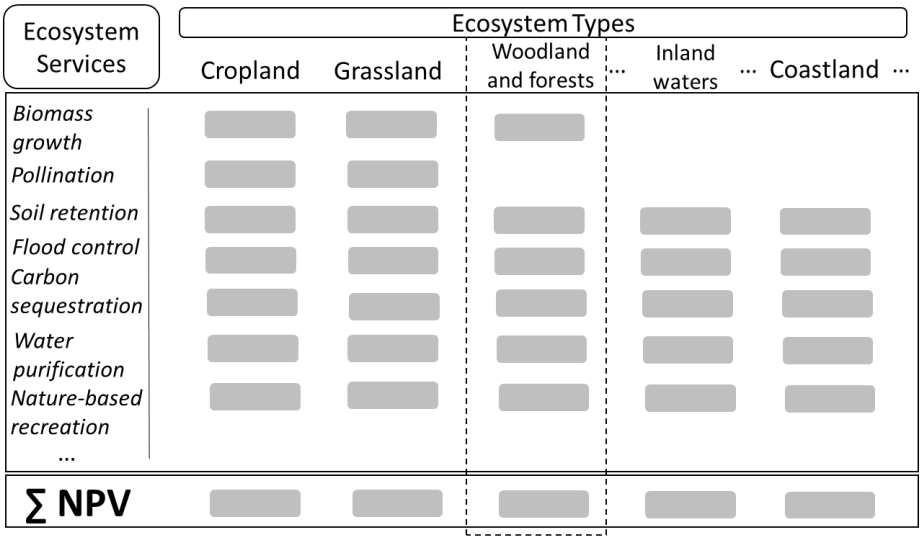
Figure 5. Simplified representation of natural resource asset accounts with a focus on the "addition" item, according to the SEEA CF



Source: own elaboration

The SEEA EA provides an asset accounting format for ecosystem extent and condition in physical terms only. For ES, the accounting format is supply and use tables, which can be in both physical and monetary terms. The ecosystem asset is accounted for in monetary terms only and is processed as Net Present Value (NPV) of ES by ecosystem type. The NPV is calculated for the year of the opening stock and again for the year of the closing stock. The difference between the opening and closing stocks is due to changes in the flows of ES provided by the ecosystem types (Figure 6).

Figure 6. Simplified representation of ecosystem asset accounts in monetary terms, according to the SEEA EA.



Source: own elaboration

Although there is a link between the provision of ES and biotic natural resources, it is not possible to calculate the ecosystem asset as the sum of the natural resource assets because the concepts are different and the underlying accounting mechanisms are different. Therefore, it is not logically correct from an accounting point of view to list forests, biodiversity and ES as three "separate" combinable entities whenever the nature nexus is to be established, because biodiversity in forests underpins the provision of many ES. An ES-based approach should, in principle, include all of them.

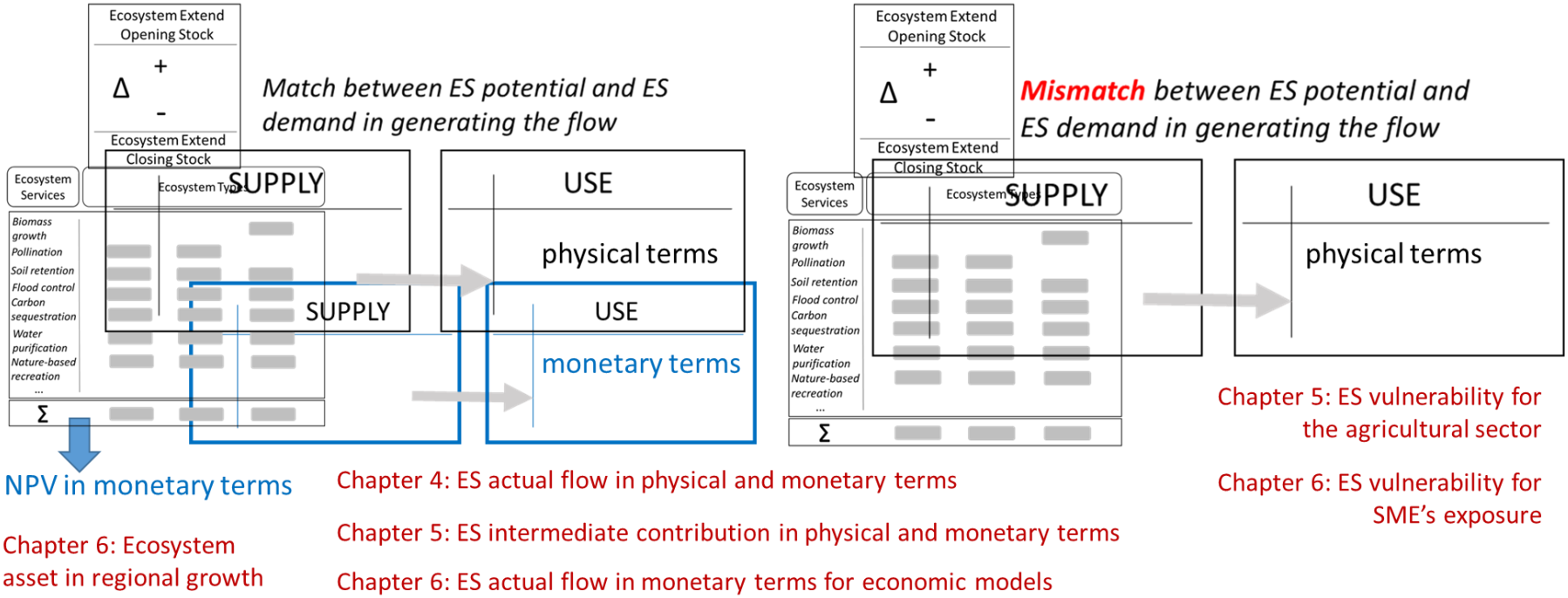
Finally, the discovery and extraction of abiotic resources cannot be considered as an ES. Although abiotic services are listed in CICES, there is no need to force the definition of ES as ecological processes to include something that is not an ecological process when looking at SEEA CF and SEEA EA in a comprehensive way:

- the SEEA CF provides formats and mechanisms for accounting for natural resources, both biotic and abiotic;
- the SEEA EA provides formats and mechanisms to account for ES. In the case of some provisioning ES, the flow may be equal to the stock addition for some natural resources;
- Abiotic services (although listed in CICES) should not be included in the ES covered by the SEEA EA.

2.3 Summary

Among the different accounting modules mentioned in this chapter, Figure 7 highlights which are the ES accounting modules, which are used in the applications presented in Chapters 4-6.

Figure 7. Simplified accounting frameworks used for each reported application.



Source: own elaboration

3 Different ways to use ES accounts and how they relate to the Sustainable Development Goals

ES accounts provide maps and tables that can be used in a variety of ways and for a variety of purposes. Their use may be direct or require additional processing. In this chapter, we consider the 3 main possible use forms to frame the applications,, namely:

1. Cases where no additional processing is needed;
2. Cases where additional processing is needed because a new indicator is created. However, the new indicator does not require any consistency with already existing tools or models;
3. Cases where additional processing is needed because ES accounts are used as inputs in existing models, tools, frameworks and therefore coherence and consistency are key.

Moreover, it is important to reconnect the use of ES accounts to the three sustainability dimensions and the internationally acknowledged Sustainable Development Goals (SDGs), which are conceptually explained in La Notte (2025a) and concretely introduced in the following section. Based on both criteria a rationale is developed to present different applications that make use of ES accounts. The chapters that follow are devoted to each of the different streams of applications.

3.1 Definition and importance of the Sustainable Development Goals

The UN 2030 Agenda for sustainable development, adopted in 2015, is a worldwide plan of action for people, planet and prosperity. Its 17 Sustainable Development Goals (SDGs) and targets aim to eradicate poverty, protect the planet, and place all countries on a sustainable development path by 2030 (UN, 2015). These goals and targets are integrated and indivisible, meaning that they are interrelated and must be considered as a whole. The SDG wedding cake (Figure 8) effectively illustrates the interconnectedness and importance of the social, economic, and ecological systems in realizing the global goals. In the three-tiered cake, each system is nested within the others, meaning that economies and societies should be perceived as embedded components of the biosphere. Such a conceptualization promotes an integrated perspective of the three dimensions of sustainability, that are biosphere, society and economy dimensions–.

Figure 8. The SDG Wedding Cake

Source: Stockholm Resilience Centre

Given the broad scope of the Agenda, it is essential to have a clear understanding of the interconnections, encompassing both synergies and trade-offs, between the diverse domains. Ecosystem accounts - measuring the contribution of nature to the economy and human wellbeing - can help to make these links explicit. They can be a useful tool to track progress towards SDG implementation, to inform policymaking and guarantee accountability (UNSD, 2020). Moreover, they can inform the entire 'policy cycle', from problem definition to policy formulation, evaluation and review, enabling sustainable policy decisions and actions needed to attain the SDGs. In this way, ecosystem accounts can support the SDGs by ensuring that natural resources are integrated in both planning and implementation process, both at the global and local level (Huang et al. 2024; Yin et al., 2021; UN, 2021; Ruijs et al. 2018; WAVES, 2016).

The pivotal role of ecosystem accounts in delivering the SDGs is explicitly articulated in one of the Agenda's targets (15.9) that aims to "integrate ecosystem and biodiversity values into national and local planning, development processes, poverty reduction strategies and accounts". The corresponding UN SDG indicator (15.9.1b) focuses on the "integration of biodiversity into national accounting and reporting systems, defined as implementation of the System of Environmental-Economic Accounting" (SEEA). By offering a multifaceted view on the interconnections between the economy and the environment, the SEEA can help reveal trade-offs and synergies across the policy domains, addressing the need for policy coherence for sustainable development (SDG 17.14) (UN, 2021).

Drawing inspiration from the SDG wedding cake visualization (Figure 8), the analysis shows that the primary level of utilization pertains to the biosphere, as descriptive statistics, without the need for

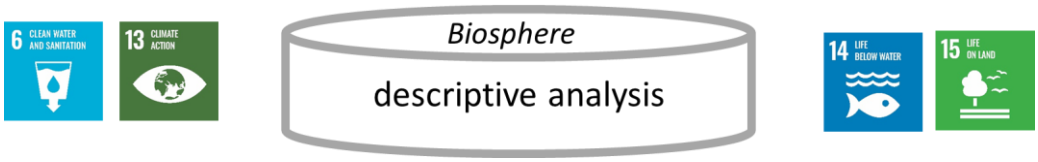
additional elaboration. This encompasses ecosystem services related to water (SDG 6), climate change (SDG 13), terrestrial and marine ecosystems (SDG 15 and 14). The subsequent level of account application pertains to the societal sphere and comprises all measurements that require the integration with other data to process novel indicators. Lastly, the third layer refers to the accounts utilised as input variables in models and tools, predominantly within the economic domain.

3.2 Use the ES accounts for conventional descriptive analysis

Statistics and accounts are a means, not an end. Information produced in a systematic way and according to a set of standards is made available to a wide range of users. Statistical offices usually provide accounts together with some basic statistics, which are used to provide a general overview of the situation and trends. The first set of uses concerns descriptive analysis, which refers to using current and historical data to identify trends and relationships. Depending on the accounts developed, it is in principle possible to carry out an initial analysis of the current state of ecosystems and ES flows and to check where increases or decreases have taken place for the available time series.

Once ES accounts are available, they will be able to provide information on which ecosystems provide support for human activities without the need for further processing. A kind of systematic reporting can be generated to provide an overview of the actual flows of ES for the area of interest. Framing them to the SDG context, ES accounts are useful for descriptive analysis in most of the targets related to the biosphere (Figure 9).

Figure 9. ES accounts to provide descriptive statistics to measure and monitor the biosphere



Source: own elaboration

Specifically:

6- Clean water and sanitation: Ensure availability and sustainable management of water and sanitation for all

> ES related to water availability, such as water flow regulation, and water purification can be used to measure and monitor the provision of clean water over time. The measurement of the ES flows (match) can be complemented by the assessment of the mismatch that can provide information about ES vulnerabilities for both water availability (unmet demand) and water purification (overuse);

13- Climate Action: Take urgent action to combat climate change and its impacts

For climate change, there are ES related to mitigation, such as carbon storage and sequestration, and ES related to adaptation, such as flood protection, coastal protection, fire protection, microclimate regulation, etc. In both cases, the measurement of ES flows (match) can be complemented by the assessment of the mismatch, which can provide information on the vulnerability of the ES (which, depending on the ES, can be due to unmet demand or missing flows);

14- Life below water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development

All marine ES can support this goal through the measurement of those ES that provide ecological inputs (such as fish provision, raw biotic material), pollution removal (such as bio-remediation by aquatic plants), protection (such as coastal protection) and overarching environmental target achievements (such as blue carbon and habitat and species maintenance). The measurement of the actual flows (match) for all these ES can be complemented by the assessment of the mismatch that can provide information about ES vulnerabilities (that depending on ES can be due to overuse, unmet demand, and missed flows);

15- Life on land: protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

All terrestrial ES can support this goal through the measurement of those ES that provide ecological inputs (such as timber provision, crop pollination), pollution removal (such as air filtration, soil de-contamination), protection (such as flood control, fire protection) and overarching environmental target achievements (such as global climate regulation and habitat and species maintenance). The measurement of the actual flows (match) for all these ES can be complemented by the assessment of the mismatch that can provide information about ES vulnerabilities (that depending on ES can be due to overuse, unmet demand, and missed flows).

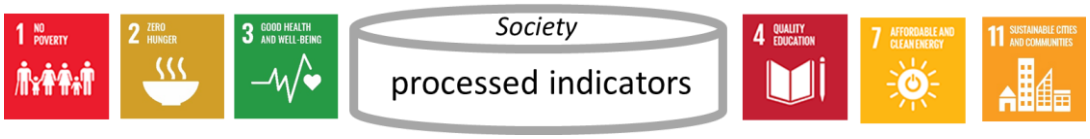
ES such as water purification, carbon sequestration, flood control may support more than one SDG (SDG 6 and SDG 15; SDG 13 and SDG14). Any assessment that requires a quantification of unsustainable patterns would need the complementary measurement of ES vulnerabilities (mismatch). Chapter 4 illustrates the application of descriptive analysis based on the INCA accounts run for the EU 27 member states.

3.3 Use of ES accounts in combination with other data to produce novel indicators

The same ES already used in descriptive analysis can be used to process new indicators. This effort may require (i) the inclusion of other data sources and (ii) the creation of ad hoc methodologies to meet the purpose of the assessment.

In the SDG framework, the new indicators including ES can capture most of the targets related to society, the second layer of the SDG wedding cake (Figure 10).

Figure 10. ES accounts to process indicators that measure the interaction with society



Source: own elaboration

Specifically

1- No poverty: end poverty in all its forms everywhere

> ES related to the protection of human settlements (such as coastal protection, flood control) are linked to the objective of increasing the resilience of people in vulnerable situations and reducing their exposure and vulnerability to climate-related extreme events and other economic, social and environmental shocks and disasters. To serve this objective, ES accounts need to be processed together with other data, related to population and hazard risks in specific areas;

2- Zero hunger: End hunger, achieve food security and improved nutrition and promote sustainable agriculture

> The resilience of the food system requires not only the environmental pillar (which can be supported by all ES that provide ecological input for the provision of food (such as fish provision, crop pollination, soil retention, water flow regulation) but also the social and economic pillars. In addition, indicators can be processed to assess not only the current path, but also whether unsustainable paths are being followed, leading to non-resilient food systems;

3- Good health and well-being: ensure healthy lives and promote well-being for all at all ages

> ES related to pollution (such as air filtration, soil de-contamination and water purification) can be important contributors to indicators that assess mortality caused by unsafe water, poisoned soil and polluted air. Indeed, the emission of pollutants is the driving force that affects the ecosystems in which people live. Targets to increase the capacity of ecosystems to provide ES sinks can have an impact on reducing illness and mortality rate;

4- Quality education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

> One specific ES can support this objective by contributing to a series of indicators: nature-based education, centred on the engagement with natural elements as a pathway for learning;

7- Affordable and clean energy: ensure access to affordable, reliable, sustainable and modern energy for all

> Biotic resources from terrestrial and marine ecosystems can act as a source of renewable energy, together with the ES related to water availability. Together with other sources of information, these ES flow can be used to process renewable resource indicators;

11- Sustainable cities and communities: make cities and human settlements inclusive, safe, resilient and sustainable

> Some ES are specifically related to the urban context (such as air filtration, micro-climate regulation) and some others can be treated differently (in terms of scale and ecological characteristics) when applied to an urban context (such as water run-off and nature-based recreation). All these ES, in their actual flows and vulnerabilities, can contribute significantly to the assessment of sustainability indicators for human settlements.

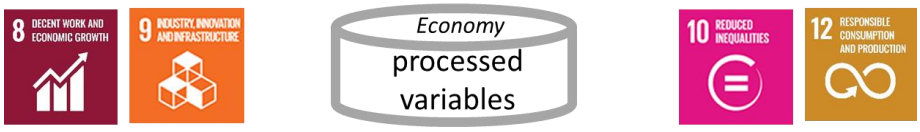
In some cases there may be a specific reference to particular ES, such as nature-based education (SDG 4); in other cases, the reference is to a group of ES according to the role they play in supporting society, such as providing ecological inputs for nutrition (SDG 2), or improving health by removing pollutants (SDG 3), or protecting vulnerable people against extreme events and natural disasters (SDG 1); in other cases, the reference is to the specific ecosystem type in which ES are provided, such as urban ecosystems (SDG 11). Chapter 5 illustrates applications to new indicators, also based on the INCA accounts.

3.4 Use ES accounts as an input variable in existing tools/ models/ frameworks

ES accounts can be used as input variables in already existing models and tools. This is particularly true for those models and tools that use the System of National Accounts (SNA) by default. As ES accounts follow the same classifications, accounting rules and principles as the SNA, their use as satellite data source in economic models is greatly facilitated. However, given the inherent complexity of ES (which remain ecological processes supporting human activities), their use requires some intermediate processing in (i) preparing the data as input variables, (ii) correctly interpreting their meaning and role as input variables.

In the SDG framework, ES accounts can be used as input variables for the SDGs related to the economy, the third layer of the SDG wedding cake (Figure 11).

Figure 11. ES accounts to provide input variables into economic and financial models, tools, frameworks



Source: own elaboration

Specifically:

8- Decent work and economic growth: promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

> ES accounts can be used in economic models that process indicators at different administrative and spatial scales;

9- Industry, innovation and infrastructures: build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

> The key ES for innovation is when nature is used as a source of inspiration for research and development. Biomimicry is the discipline and "mimicking nature" is the service (REF) that can (and should) be included in the innovation-related economic analysis;

10- Reduce inequality: reduce inequality within and among countries

> ES accounts can be used to build financial soundness indicators, to ensure equity in the markets, to integrate as appropriate information about the economic dependency on ES. Financial soundness indicators, in fact, are specific indicators identified for SDG 10 to improve the regulation and monitoring of global financial markets and institutions;

12- Ensure sustainable consumption and production patterns

> ES accounts can be used in multi-regional input-output frameworks to process footprint-like indicators that show ES dependency not only in production but also in consumption through trade in goods and services.

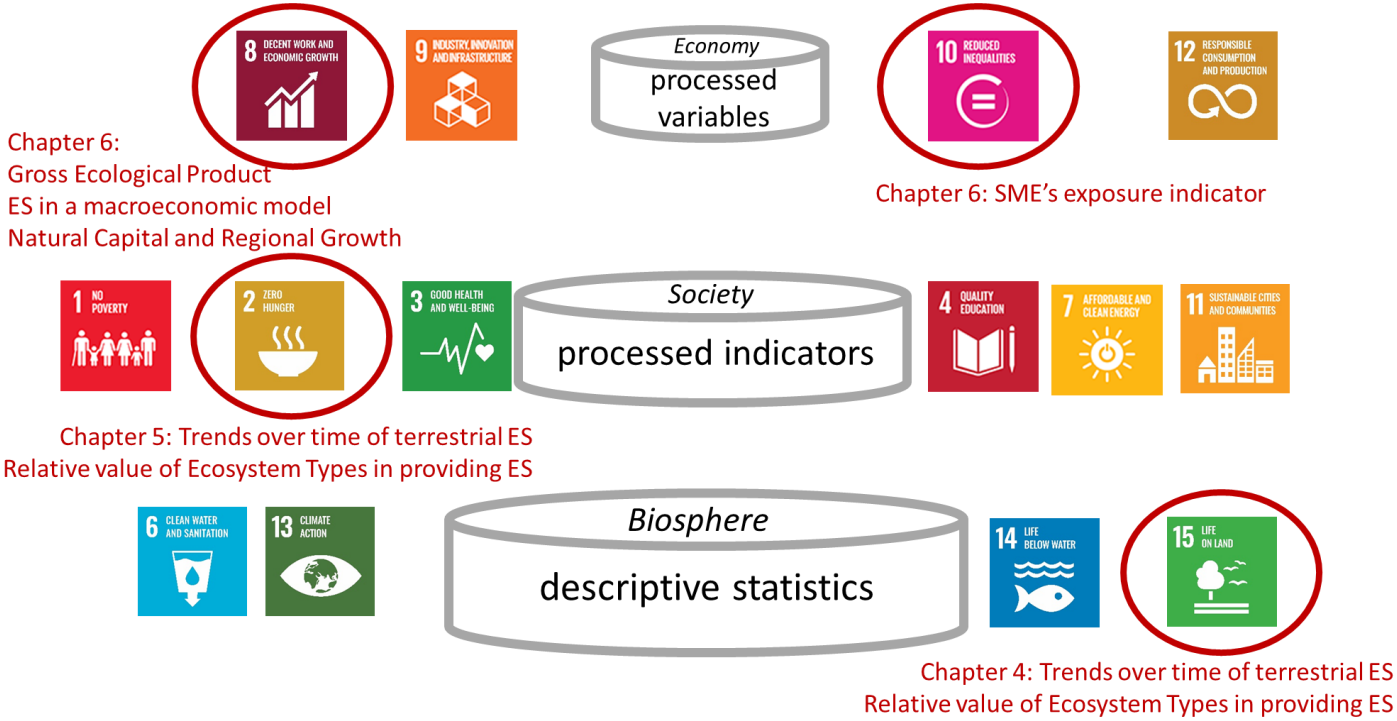
The basic characteristic of this third group is that ES are mainstreamed into economic frameworks. Although some ecological concepts and contents are new, the context is and remains "economic" because the conventional economic tools and models are used to carry out the analysis. Chapter 6 illustrates applications to economic indicators that have been adapted, integrated, or reformulated by using the INCA accounts.

3.5 Summary

Figure 12 provides an overall picture of the SDGs linked to the applications presented throughout the report. The examples reported in Figure 12 refer only to what described in the next chapters, but the potential use of ES accounts can potentially cover all the SDGs, as previously explained.

Moreover, some clarifications are needed. First, the focus is on ES accounts in particular, and not on SEEA in general. This means that accounts related to (for example) environmental expenditure, which can be linked to the SDG target 15.a ("Mobilize and significantly increase financial resources to conserve and sustainable use biodiversity and ecosystems"), are not reported, nor are accounts related to the extent of ecosystems. Second, in some cases the ES match univocally one SDG target, while in other cases the ES accounts are useful to monitor the SDG as a whole. Third, the way ES accounts are grouped according to the SDG layers as reported by REF is purely indicative, as there may be much overlap.

Figure 12. Connection between the uses of Ecosystem Services accounts and the Sustainable Development Goals: some examples

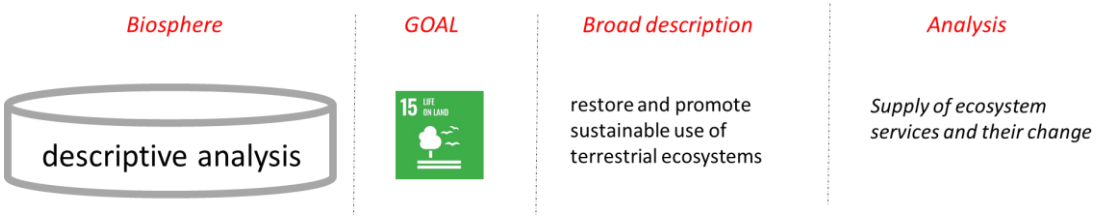


Source: own elaboration

4 Ecosystem Services accounts in descriptive analysis

The purpose of ES accounts is to measure ES flow and monitor it over time. Compilers of ES accounts can directly perform these tasks by using the accounting tables and undertake the necessary analyses. This chapter presents the descriptive analysis of the INCA accounts on ES in two different policy contexts. First, the European data are examined with the aim of analysing the situation on the continent; second, the ES accounts are used for a headline indicator in a global framework. The two cases show that the same data source can provide multiple opportunities, even without further processing. As the INCA data refer only to terrestrial ecosystems, the focus is on the SDG 15 "Life on land" in general, since the purpose is to measure and monitor ES over time (Figure 13).

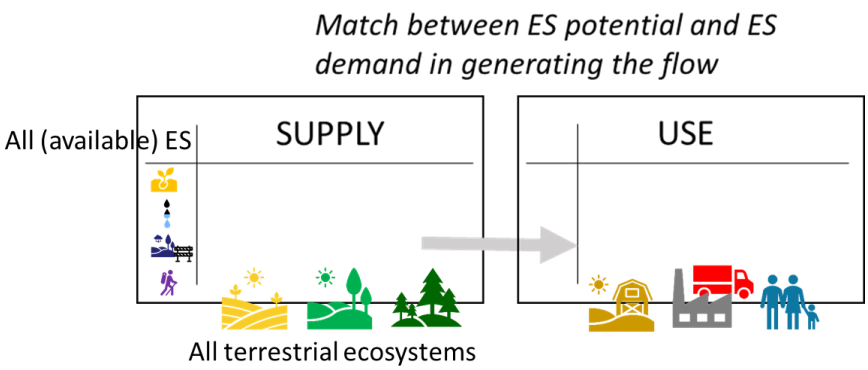
Figure 13. SDG: goal, broad description and kind of analysis for the cases of basic descriptive analysis



Source: own elaboration

In both cases, the ES accounts are about the Supply and Use Tables (SUT) of the actual flow, i.e. the ES flow that is effectively delivered. Therefore, neither asset accounts nor mismatch flow accounts are necessary (Figure 14).

Figure 14. SDG: goal, broad description and data for the cases of basic descriptive analysis



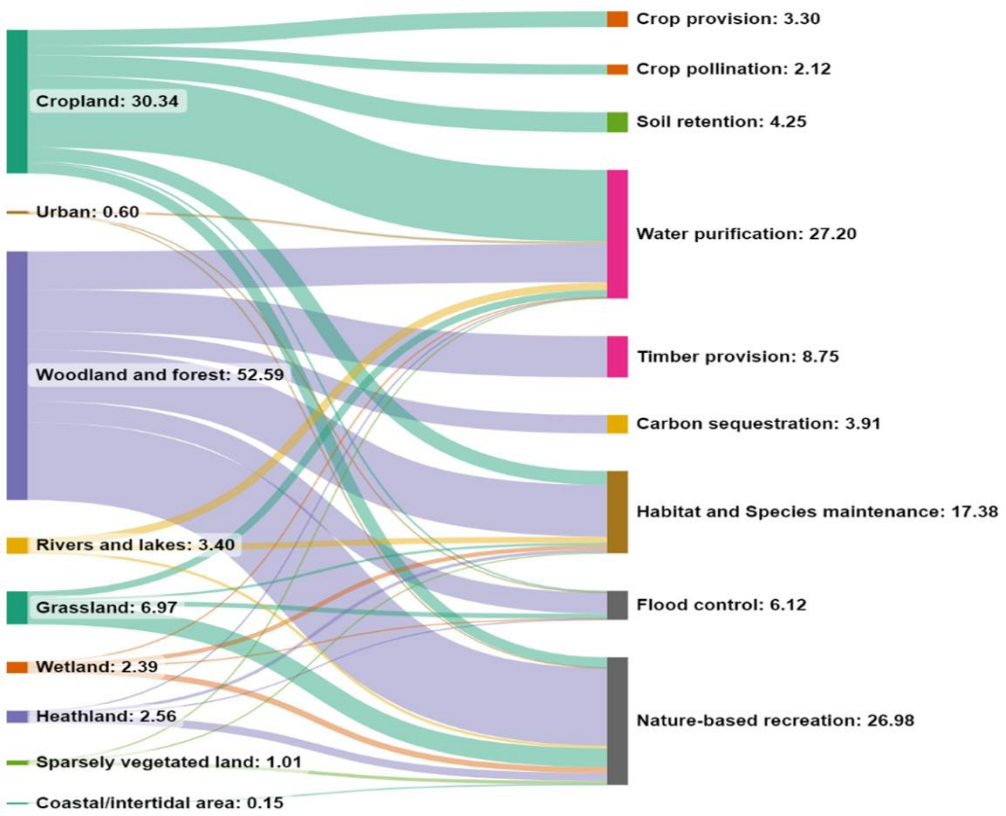
Source: own elaboration

The two applications presented in this chapter provide concrete examples of how the ES accounts can be used in a straightforward manner once the ES accounts become available.

4.1 Descriptive analysis of ES in Europe

In approaching nature-related risks, a fundamental role is attributed to nature in enabling economic activities through services. If disrupted, it has the potential to significantly affect the economy and ultimately undermine financial stability. It is therefore important to measure the ES provided by ecosystems in support of the socio-economic system (Cambridge Institute for Sustainability Leadership, 2021; Network for Greening the Financial System, 2023). Specifically, the accounting module on ES offers a comprehensive view of how ecosystems support specific sectors and beneficiaries Based on the nine ES accounts produced under INCA for four accounting periods - 2000, 2006, 2012, and 2018, an overview can be provided of the current state of ecosystems through the ES supply (Figure 15). Results from the INCA project (European Systemic Risk Board, 2023) suggest that the flow of nine ecosystem services in accounting year 2018 in EU-27 context, has an estimated value of 187€ billion (in 2012 prices). In absolute terms, forest and woodland provide almost half of the total value, followed by cropland (30%). In terms of ES, water purification and nature-based recreation represent each almost 27% of the total value, followed by habitat and species maintenance service (17%).

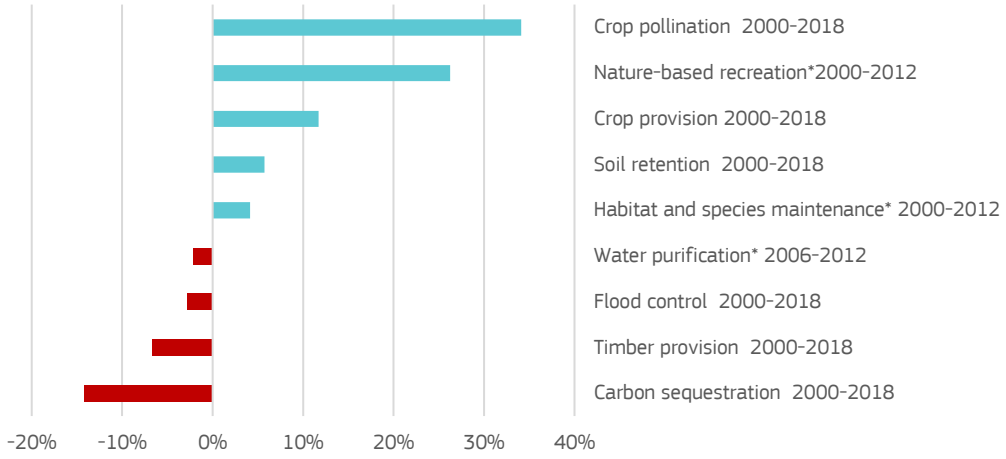
Figure 15. ES supply in % of the aggregated value in monetary terms (2018)



Source: ESRB/ECB (2023)

Looking at changes over time (Figure 16), the results show an increase of nature-based recreation (+20%), crop pollination (+30%) and crop provision (+10%). On the other hand, there is a decrease for water purification, flood control, timber provision and carbon sequestration services. Carbon sequestration, followed by timber provision, has the largest negative change due to land transformation.

Figure 16. Percentage change in ES actual flow between the reported accounting periods (*) in physical terms



Source: adapted from ESRB/ECB (2023)

It is important to note that what is reported in Figure 15 and 16 only considers a small number of ecosystem services (nine out of the ninety ES currently listed in CICES v.5.1[5]) made available by the INCA project.

4.2 The GBF Headline indicator B.1: Ecosystem Services

The Kunming-Montreal Global Biodiversity Framework (GBF), agreed by parties to the UN Convention on Biological Diversity (CBD, 2022), set a vision, four goals and 23 targets for biodiversity conservation for 2030. As a Party to the Convention, the primary multi-lateral treaty on nature, the EU has committed to the achieving GBF, and to reporting on progress using headline indicators outlined in its monitoring framework. Goal B of the GBF outlines a long-term goal for ecosystem services:

> *"Goal B. Biodiversity is sustainably used and managed and nature's contributions to people, including ecosystem functions and services, are valued, maintained and enhanced, with*

[5] Ref. https://cices.eu/

those currently in decline being restored, supporting the achievement of sustainable development for the benefit of present and future generations by 2050."

The headline indicator for Goal B is B.1 Services provided by ecosystems. Headline indicators are intended as mandatory for national reporting. As was the case for 11 headline indicators, at the time of adoption, there was no existing methodology, and an Ad Hoc Technical Expert Working Group (AHTEG) on indicators was established to guide the development of these indicators, as part of their mandate to provide guidance on the methodologies for all the headline and binary indicators (Affinato et al. 2024). A task team was set up by the UN Statistics Division, the assigned custodian of the indicator, to develop and test a methodology for B.1 that could support monitoring of progress towards Goal B.

The aim of the development and testing process for B.1 was to identify and develop the most suitable indices that can act as reliable indicators using ecosystem service (ES) time-series, to improve understanding of the options for representing trends in ecosystem service provision, and monitoring progress towards Goal B. This included evaluating different mathematical formulations for aggregating data types, using a combination of theory and practical application.

The task team proposed an index based on average trends in ES, like other biodiversity indicators (such as the Living Planet Index (Collen et al. 2009), which gives an average in abundance trends for vertebrate animals). This solves the 'apples and oranges' problem, of combining trends in services measured in biophysical terms, using different using different units that are not comparable (as opposed to monetary values, which are assumed to be fungible). Biophysical units are converted to a proportional change in that service based on a baseline year, and the average proportional change across ES is reported. Ecosystem accounts, which can encompass both physical and monetary metrics, provide a structured and standardized framework for capturing the contributions of ecosystems to economies and society. The UN standard for ecosystem accounting in the UN System for Environmental Economic Accounting – Ecosystem accounting (SEEA EA; UNCEEA, 2021).

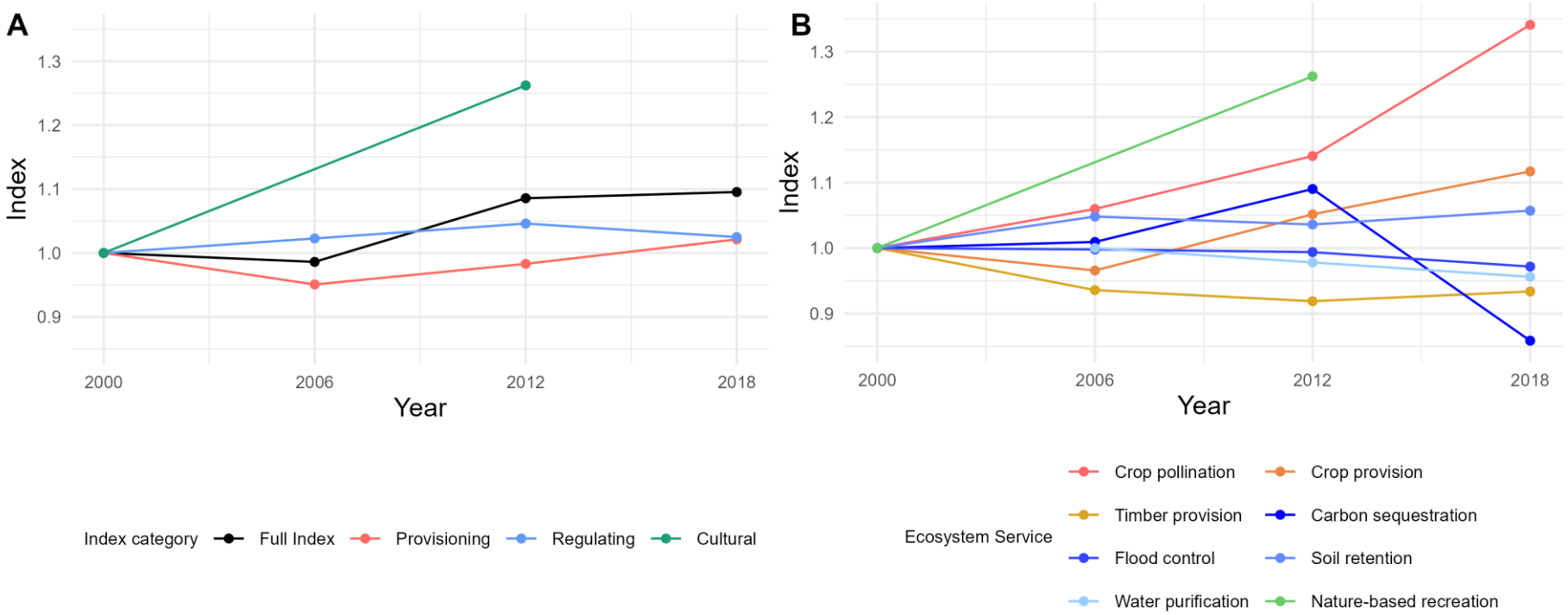
The proposed methodology for the B.1 indicator is to present an average of ES trends, using a geometric mean to average trends. When dealing with proportional change, the most appropriate means of averaging data is the geometric mean. The geometric mean accounts for proportional sensitivity by weighting high rate changes more heavily and diminishes the skewing effect of extreme values. We used a chain method, like Collen et al. (2009), to allow for different starting dates per ES time-series.

In adition to presenting an overall ES index, the taskteam recommended that countries should also present an index for each ES category (as defined in the UN SEEA ecosystem guidelines):

1. Provisioning services: material products directly obtained from ecosystems; examples include food, water, and wood.
2. Regulating services: benefits from the regulation of ecosystem processes; examples include air quality, climate regulation, and water purification.
3. Cultural services: non-material benefits people gain from ecosystems through their interactions with nature; examples include recreation, aesthetic experiences, and cultural connection.

In the EU from 2000 to 2018, on average measured ES provision increased by 5%. Provisioning and regulating services remained relatively stable, with the former increasing provision by 2% between 2000 and 2018 and the later decreasing by 2%. Cultural services were only measured at two points, 2000 and 2012, but rose by 18% between these years.

Figure 17. The B.1 index values for EU data: A) overall Index (all ES types, in black, n=8), Provisioning (red, n=2), Regulating and Maintenance (blue, n=5), and Cultural services (green, n=1).; B) Trends for ES types are coloured by Provisioning (red-orange), Regulating and Maintenance (blues), and Cultural (greens).



Source: own elaboration

By participating in the development and testing of the B.1 indicator, the EU ecosystem accounts provided insights into how ES can be measured, aggregated and their trends understood, with the context of national, EU-wide and global goals for conservation and sustainable use of the natural world. The results of the testing and development by the Task Team were presented to and reviewed by the AHTEG. The resultant meta-data were presented to Parties to the Convention at SBSTTA-26 and COP16 of the UN Convention on Biological Diversity, and subjected to review by countries between SBSTTA26 and COP16.

The application of EU data to this testing does however raise questions about the reporting burden for countries as they develop their accounts and B.1 indicator. Further development of global and regional datasets that can be downscaled to national levels, and globally available and accepted methods to enable ease of ES estimation by countries may ease this burden. Lessons learnt from the EU project may also support capacity building in other countries, facilitating the development of accounts and data-streams for the B.1 indicator.

5 Ecosystem Services accounts used to process new indicators

Maps and tables related to ES can be a valuable source of information for processing a set of indicators that are not thought to fit into a specific model or tool and therefore maximum flexibility in terms of interaction with other data sources. This chapter presents two examples of indicators that specifically address the sustainability of the food system. The first indicator relates to the dependence of the fish supply on seagrass, and specifically to the assessment of how much of the current fish landings for selected species could be at risk if the whole food web loses a fundamental component. The second indicator relates to the exposure of the agricultural sector to ES vulnerabilities with respect to the dependency on nature for what concerns the provision of ecological inputs, the removal of pollution, and the protection against natural disasters. For each application a synthetic explanation of the indicator is provided after:

- the reference to a specific SDG and target is made explicit, and
- the reference to the specific accounting module is specified.

5.1 Dependence of fish supply on seagrass

Seagrasses play a pivotal role in the marine ecosystem, exerting a profound impact on the structure and functioning of the food web. Seagrasses are a food source for many herbivorous animals, such as invertebrates, sea turtles, manatees, fish that in turn serve as food sources for other fish (Cebrian et al 1997, Darnaude et al. 2022). Therefore, the role of seagrasses extends beyond the ES that they provide directly, for example, they generate biomass for industrial processing and the sequestration and storage of carbon. Indeed, their presence (or absence) also greatly affects the provision of many more ES. This is demonstrated by the findings of a case study conducted in the Mediterranean Sea (Addamo et al., 2025), which are presented here to illustrate the application of the seagrass dependency indicator.

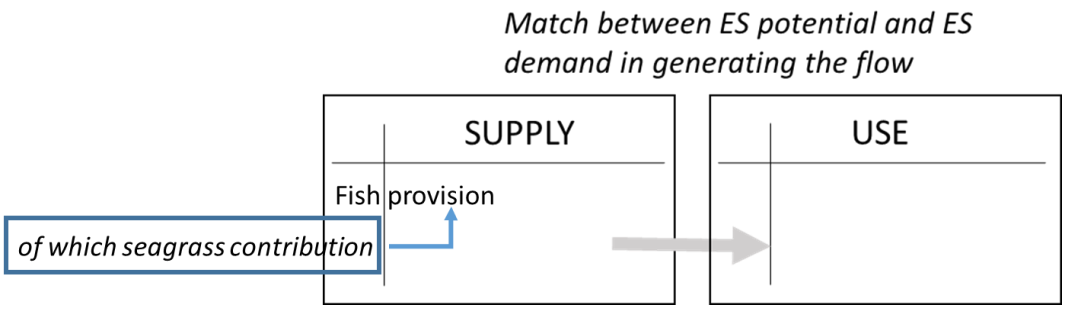
The SDG that could be best supported is SDG 2 concerning food security, and particularly the Target (2.4) related to sustainable food production systems and practices that increase productivity and production. The role of seagrass connects to the social dimension because in some countries the subsistence of entire population depends on fishing for own consumption. To destroy or damage seagrass and in turn affect the presence and abundance of fish could become in the medium and long terms the end of a critical source of food for people (Figure 18).

Figure 18. SDG: goal, target and novel indicator for the seagrass dependency of fish provision

Source: own elaboration

The accounting modules that support the calculation of this indicator refers to the actual flow of the fish provision ES. However, the key information that is processed out the total actual flow is the seagrass contribution through the food-web network for each selected fish species (Figure 19).

Figure 19. Processed information from Supply and Use Tables: ES match



Source: own elaboration

The geographical boundaries of the assessment undertaken in Addamo et al (2025) are the Geographical Sub Areas (GSAs) identified by the Food and Agriculture Organization (FAO) in the General Fisheries Commission for the Mediterranean area (GFCM). Fish provision (excluding aquaculture activities) refers to the provision of fish biomass for human consumption and is measured considering total landings or catch per unit effort (in tonne, t) for each target species. Addamo et al (2025) compute in year 2018 data retrieved from the FishStatJ software v.2021 of FAO Regional capture fisheries statistics, cross-checked with ICCAT statistical databases[6] for tuna, and STECF reports for all the other species, that are European anchovy, striped seabream, red mullet, blackspot seabream, common pandora, and sardina pilchardus.

The trophic-based seagrass dependency (TB_SGd) represents the contribution to the total mass of living organisms in each trophic level. This is calculated by considering two factors: firstly, the food-web network and secondly, the different functional groups (FGs) included in each trophic level (TL) linked to seagrass. In accordance with the methodology proposed by Piroddi et al. (2022), both the biomass and diet composition matrix for the Mediterranean marine ecosystem have been employed to ascertain the proportion of prey-predator biomass for the entire trophic web, extending from seagrass to each fish target species included in this study. The total TB_SGd represents the overall contribution of seagrass as a primary producer (TL1) to the final trophic biomass, which encompasses species from the lower trophic levels (e.g., prey TL2) to the higher trophic levels (e.g., top predators, TL5). TB-SGd is calculated by multiplying the diet contribution of each prey-predator pair involved in all TLs and FGs of the trophic web linked to seagrass.

[6] Ref. https://www.iccat.int/en/accesingdb.html

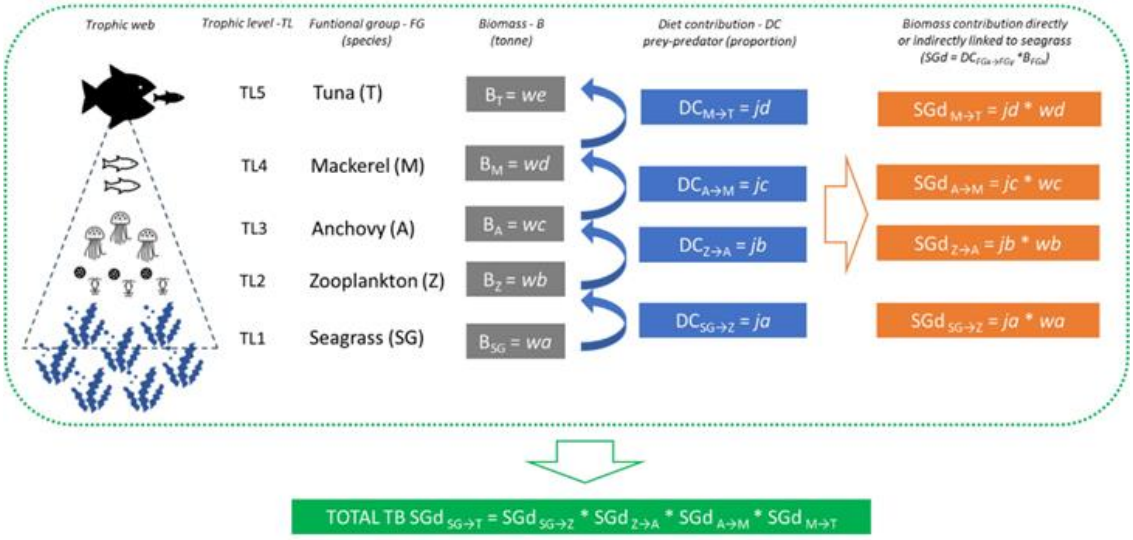
Equation 1. The trophic-based seagrass dependency

$$TB_SGd = \prod DC_{FG_x \rightarrow FG_y} * B_{FG_x}$$

Where:
$DC_{FGx->FGy}$: diet contribution (%) of prey FG_x to predator FG_y
B_{FGx}: biomass (t) of prey FGx

As shown in Figure 20, TB_SGd enables to quantify the role of seagrass in the provision of the selected fish species.

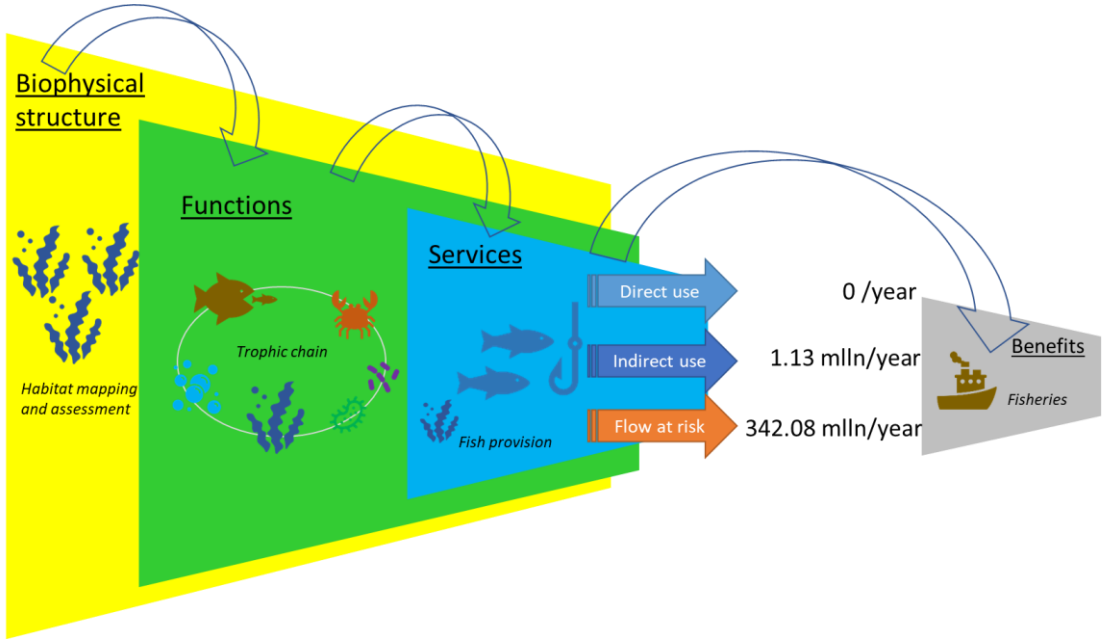
Figure 20. Trophic-based seagrass dependency (TB_SGd)



Source: Addamo et al. (2025)

After the monetary valuation based on the resource rent approach (Addamo et al., 2025), it is possible to extract for the selected fish species, the share that directly depends on seagrass habitats, and the rest of the flow that is put at risk if seagrass habitat was not there as a primary producer TL (Figure 21).

Figure 21. Direct and indirect use of seagrass for fish provision and the complementary value that could be at risk if seagrass (as primary trophic level) were not there (year 2018, EU member states bordering the Mediterranean Sea for 5 selected fish species)



Source: own elaboration

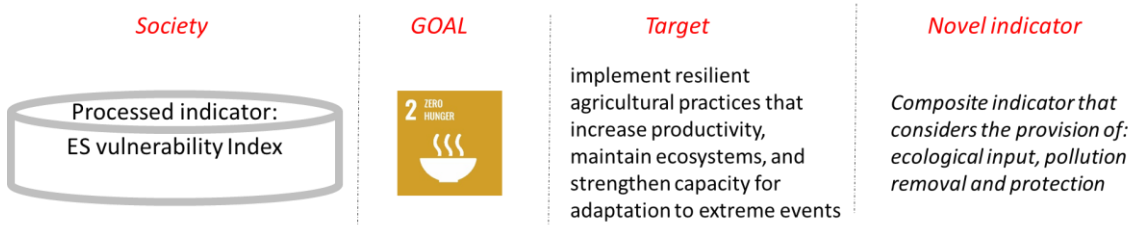
On the one hand, seagrass does not provide any direct use value to fish provision. Furthermore, the contribution computed across all the TL appears to be proportionally very low when compared to the final value of the fish provision ES (approximately 0.3%). However, as the primary trophic level, its degradation and loss may jeopardise the entire trophic chain, thereby undermining the value of fish provision as a whole. In considering the strategic importance of seagrass for fish provision in the long term (with reference to a resilient food system perspective), it is evident that the value of seagrass extends beyond the 1.13 million/year value. Rather, it encompasses the remainder of the fish provision flow that would otherwise be jeopardised.

5.2 Ecosystem Services Vulnerability Index for the agricultural sector

Ecosystem services can contribute to the economy and society in many ways. On the one hand, they can come as a direct contribution in the form of ecological inputs to production processes (e.g. growth of food and non-food crops, wild pollinators), remove negative externalities from production (e.g. air filtration, water purification), or protect activities from physical and biological hazards (e.g. flood control, pest regulation). On the other hand, where there are overarching, internationally recognised environmental objectives (such as mitigating climate change and halting biodiversity loss), and the proper management of ecosystems contributes to the pursuit of these objectives (e.g. carbon sequestration and biodiversity conservation). Within the primary sector, agriculture relies heavily on ES as a provider of ecological inputs, as a purifier of negative externalities generated during the production process, and to protect the large area of fields from natural disasters. The resilience of the whole food system, which in turn depends on agriculture, therefore, depends on the provision of ES. If such provision is not ensured, the sector is exposed to ES vulnerabilities that can

be measured. The SDG that could be best supported by such measurements is again SDG 2 related to food security and, in particular, the Target (2.4) related to sustainable food production systems and practices that increase productivity and production.

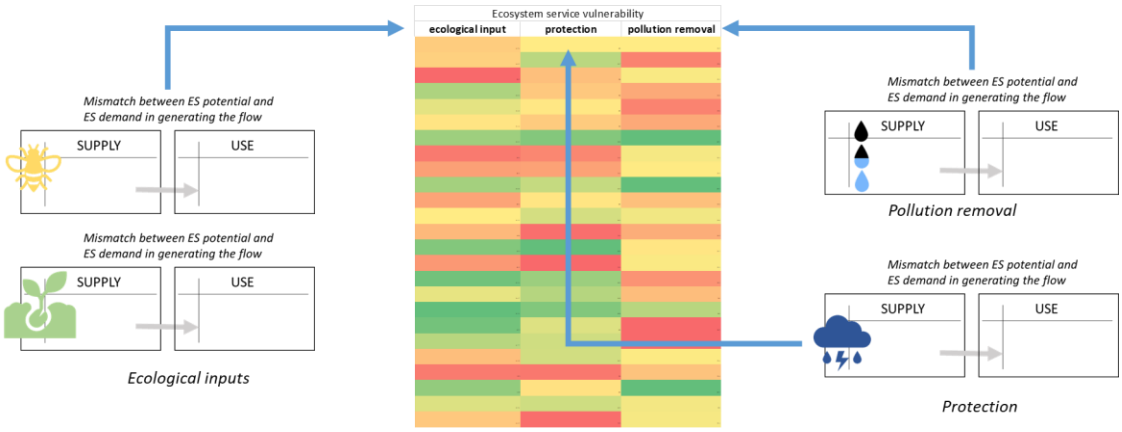
Figure 22. SDG: goal, target and novel indicator for ES Vulnerability Index for the agricultural sector



Source: own elaboration

Socio-economic demand can exceed the ecological supply, thus resulting in a mismatch. Mismatch accounts where ES potential < ED demand are the most effective metrics to measure nature-related risks and are named as "ES vulnerability accounts" (La Notte et al. 2025b). This can be the case (i) if an ecosystem service is used above its sustainability thresholds (e.g. the extraction of natural resources above their regeneration rates, or the emissions of pollutants above ecosystems' capacity to absorb them), or (ii) due to the absence of ecosystems that are capable to provide the services needed (e.g. lack of vegetated land to control the risk of erosion, lack of habitats suitable to host wild pollinators). Average mismatch values are calculated for the main groups of ecosystem services accounted in INCA with a focus on the agricultural sector as an initial application, although this approach can be replicated also for other sectors. Specifically, (i) the group "ecological inputs" includes pollination and soil retention, (ii) the group "pollution removal" includes water purification, and (iii) the group "protection" includes flood control (Figure 23).

Figure 23. Processed information from Supply and Use Tables: ES mismatch



Source: own elaboration

ES vulnerability accounts facilitate the calculation of metrics that can be utilised to evaluate the vulnerability of economic sectors that are highly dependent on ecosystems. Specifically, La Notte et al. (2025) examined the dependency of the agricultural sector in the EU for the pilot year 2012 and sought to calculate an ES vulnerability index. For each group of ES the ES vulnerability accounts are scaled in accordance with Equation 2:

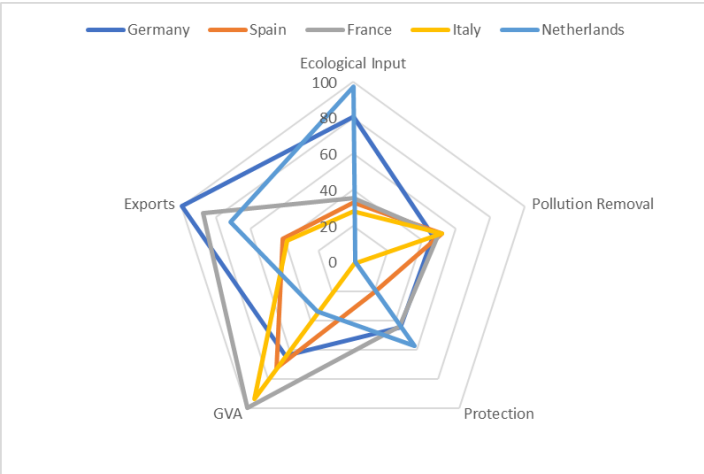
Equation 2. The ES Vulnerability Index

$$V_j = x_i - \bar{x}$$

Where,
j=1, 2, 3 i.e. 1 corresponds to "ecological inputs"; 2 corresponds to "pollution removal"; and 3 corresponds to "protection";
i=1, ..., n is the number of ES vulnerability accounts processed per pixel;
$\bar{x}$ is the average of pixel values.

The composite indicator is then calculated using the geometric mean as the aggregation rule, following normalisation (using the min-max method) and equal weighting. A comparison and contrast of ES vulnerability indicators in the EU Member States with the highest agricultural Gross Value Added and the highest agricultural exports would be an interesting exercise. France is the leading producer and exporter of agricultural products among the EU Member States, with Germany and Spain ranking second and third, respectively. Italy has a strong record of agricultural production, while the Netherlands has a strong record of exports.

Figure 24. Components of the ES Vulnerability Index in the EU Member States with the highest Gross Value Added and Exports in agricultural products



Source: own elaboration based on La Notte et al. (2025)

Based on the results presented in La Notte et al. (2025), Figure 24 shows that Italy has been identified as the most vulnerable EU Member State in terms of ecosystem protection, with Spain ranking second. France, Germany and the Netherlands are the countries that export the most agricultural products. Germany and the Netherlands are not vulnerable in terms of ecological inputs, but the Netherlands is highly vulnerable in terms of pollution removal, followed by France and Germany. These vulnerabilities expose the agricultural sector in these countries to a variety of risks. The ES vulnerability accounts are a valuable tool for measuring the shortcomings of the ecosystems on which the agricultural sector depends.

6 Ecosystem Services accounts as input data into economic models and tools

Maps and tables related to ES can be a valuable source of information as input variables to already existing models and tools that are conventionally used by researchers and analysts in the economic and financial fields. The fact that models and tools already exist places several constraints and requirements on the ES accounts that are used. Indeed, there is a strong need for coherence to ensure that the results are consistent and reliable. This chapter provides four examples of how ES accounts can be used as input variables within economic analysis frameworks. The chapter starts with a first application that considers the contribution of ES mismatches to assessing the risk exposure of small and medium-sized enterprises (SMEs), to move towards a broader conceptualisation of how to frame the contribution of ES to total factor productivity in macroeconomic models. Keeping the macroeconomic perspective, the following applications focus first on a sectoral analysis by calculating a macro indicator, and then on a spatial analysis addressing regional growth. For each application a synthetic explanation of the indicator is provided after:

- the reference to a specific SDG and target is made explicit, and
- the reference to the specific accounting module is specified.

6.1 Dependency of firms on ecosystems services

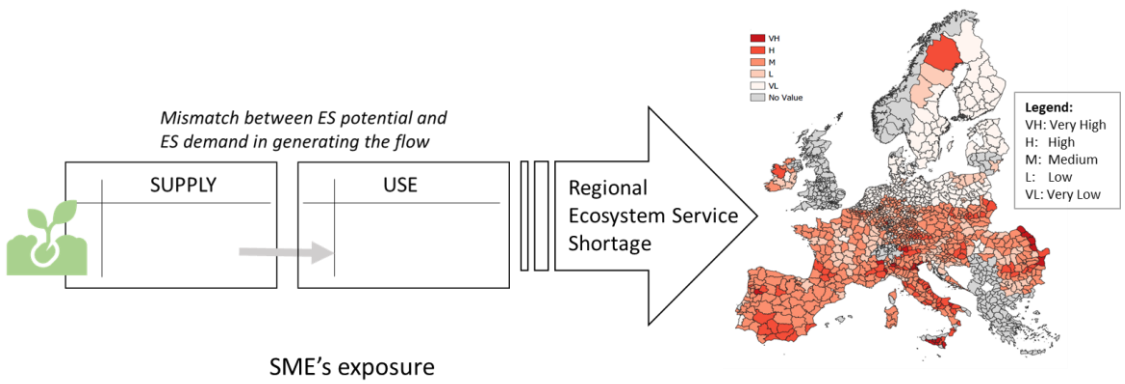
Nature-related financial risks faced by companies have become a pressing concern for policymakers and financial actors. Ecosystem services, which underpin various production processes, for instance as direct physical inputs (e.g. water) for production or protection from natural hazards such as floods, are increasingly vulnerable to degradation and overuse. This can lead to production disruptions, increased costs, and ultimately entail significant risks for economic growth and financial stability, with potential consequences for SDG 10 on inequalities. The most closely related target (10.5) refers to improving regulation and monitoring of global financial markets and institutions and strengthening the implementation of such regulations.

Figure 25. SDG: goal, target and input variable for SME's exposure to ES

Source: own elaboration

To advance modelling, it is vital to link data on economic activities, their reliance on ecosystem services and actual exposure to geospatial ecosystem services conditions. The study of Fatica et al. (2024) proposes an approach that provides a comprehensive view on nature-related financial risk indicators by combining ENCORE (see Natural Capital Finance Alliance, 2022) sectoral exposure with INCA based ecosystem services reflecting regional ecosystem services conditions. The method is illustrated focusing on European SMEs' long-term debt, which proxies for loans held by European banks. As such, the exercise provides insights on firms' vulnerabilities and their possible spill overs to the financial sector. To develop a geospatial indicator of ecosystem shortage, the study utilizes available data to calculate the demand and unmet demand for ecosystem services at the NUTS3 territorial level (Figure 26).

Figure 26. Input variable from Supply and Use Tables: ES mismatch



Source: own elaboration based on Fatica et al. (2024)

Thus, a measure of regional ecosystem provisioning shortage (EPS) is derived, as expressed in Equation 3. Importantly, this metric can be adapted and applied to assess the shortage of any other ecosystem services.

Equation 3. The Ecosystem Provisioning Shortage

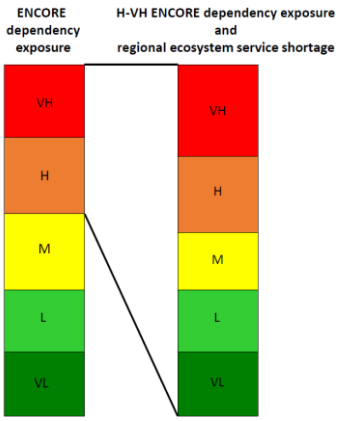
$$\text{EPS} = \frac{Demand}{Demand - Unmet\ demand}$$

The framework is operationalized by establishing severity thresholds that categorize outcomes into five distinct materiality ratings: Very Low (VL), Low (L), Medium (M), High (H), and Very High (VH). This threshold-based approach allows one to group results in a meaningful and structured manner. For example, Figure 26 illustrates the varying severity levels of regional soil retention shortages using 2018 soil retention data at NUTS3 level.

As a first step, a creditor can aggregate long-term debt positions of its debtor firms (portfolio of firms) and decide on the ecosystem dependency exposure of this portfolio grouping the individual debt positions into the materiality ratings as proposed by ENCORE. ENCORE is a holistic mapping between economic sectors (e.g. ISIC 4-digit) and their reliance on ecosystem services for their production. A higher reliance on an ecosystem service translates to a higher materiality rating. In

particular, “High” and “Very High” ratings indicate that the ecosystem service is essential to production process, and insufficient ecosystem service supply would interrupt it.

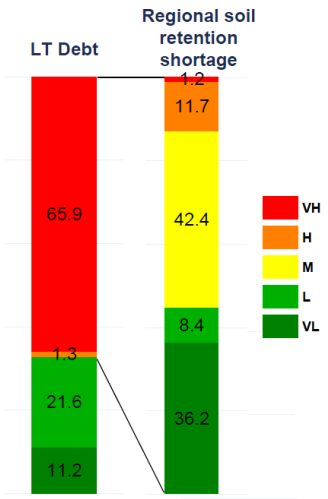
Figure 27. Scheme of the basic method to refine sector-based metrics with geospatial data



Source: Fatica et al. (2024)

The approach in Fatica et al. (2024) classifies a long-term debt portfolio of firms by its sectoral dependency exposure (Fig 27. left column), and then decomposes the very vulnerable-classified, H and VH exposed, long-term debt by the geospatial shortage metric, e.g. soil retention (Figure 27 right column). This allows to refine the pure sectoral exposures with the actual regionally ecosystem service shortage. In other words, the refined metric captures both sector and location-specific vulnerabilities. We illustrate this concept in Figure 28, which displays the case of soil retention for a portfolio of agricultural SMEs long-term debt positions. For instance, the sectoral exposure points to 67.2% of long-term debt being exposed to H and VH soil erosion risk, in reality, only 12.9% of that is really suffering a regional shortage. In other words, the refined metric captures both sector and location-specific vulnerabilities. We illustrate this concept in Figure 28, which displays the case of soil retention for a portfolio of agricultural SMEs long-term debt positions. Figure 28 displays this for the case of soil retention for a portfolio of agricultural SMEs. For instance, the sectoral exposure points to 67.2% of long-term debt being exposed to H and VH soil erosion risk, in reality, only 12.9% of that is really suffering a regional shortage.

Figure 28. Sectoral and geospatially refined risk exposure



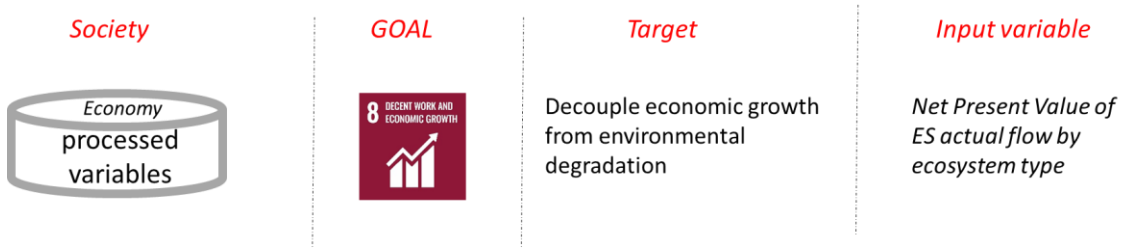
Source: Fatica et al. (2024)

As discussed above, the framework in Fatica et al. (2024) can be generalised and applied to any other ecosystem services, such as water provision, flood control, etc., to assess vulnerabilities across two dimensions – the sector of activity and the physical location of firms. Likewise, with access to firm-level balance sheet information, extension to evaluate potential fragilities in terms of other aspects of business operations, such as employment, and investment, is straightforward. With the increase in temperature and environmental degradation exacerbating climate and nature-related risks in an uneven way across the EU (EEA, 2024), the approach can inform policymaking and act as an early warning for possible pockets of vulnerability arising in areas where some economic activities are concentrated. Extending the framework entails accounting for upstream and downstream linkages among firms to account for the complexity of the geography of production and input sourcing. This is particularly compelling for SMEs, which make up over 99% of EU companies, and are already significantly vulnerable to sectoral shocks and to disruptions along the supply chain than larger companies (Fatica et al., 2022).

6.2 Ecosystem services in a simple macroeconomic framework

Although classical economists considered land as a production factor, standard contemporary macroeconomic models continue to abstract from the interactions between natural assets and the economy (for a discussion, see for example Dasgupta (2021)). In this section, we investigate the dependence of the economy on natural resources through the lenses of macroeconomic models. Concretely, we use forest ecosystem services (ES) to demonstrate the suitability of an aggregate macroeconomic production function extended to natural capital for assessing how different simulated degradation paths for forest ecosystems could impact economic production possibilities. Our work can therefore serve as proof of concept for including the contribution of natural capital more systematically in macroeconomic framework underpinning monitoring and surveillance assessments, which can address the SDG 8 related to economic growth and in particular the target (8.4) on decoupling economic growth from environmental degradation (Figure 29).

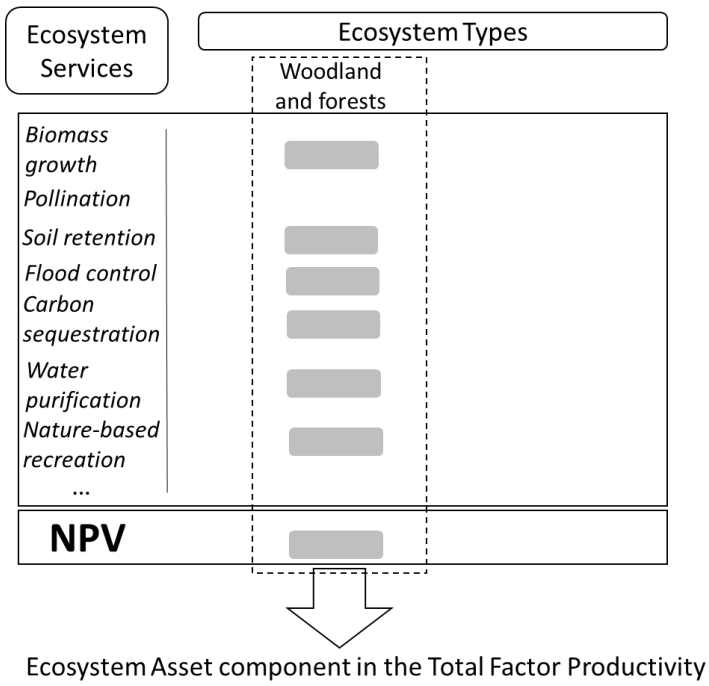
Figure 29. SDG: goal, target and input variable for introducing ES into the macroeconomic framework



Source: own elaboration

As mentioned earlier, we focus on ES provided by forests. Forests serve as a natural example for demonstrating how ES can be integrated into a macroeconomic model for two main reasons. First, they represent a significant share of ES in Europe. Second, forests offer both marketed services, such as timber, and non-marketed services, such as soil retention. This dual nature allows us to illustrate how a simple framework can differentiate between various types of services provided by natural assets (Figure 30).

Figure 30. Input variable from Supply and Use Tables: the Net Present Value of ES actual flow



Source: own elaboration

To incorporate these services into a macroeconomic production function, it is essential to focus on stocks rather than flows. Accordingly, we first calculate the net present value of ecosystem services provided by forests (specifically wood provision, climate regulation, flood control, water purification, habitat and species maintenance, and nature-based recreation), using a discount rate of 4%. We then aggregate these stock values at the country level, as the analysis is conducted for each EU member state. By applying the same discount rate used by the World Bank[7], we can directly compare our estimates for forest capital stocks with those in the Changing Wealth of Nations database. Finally, we utilise the constructed series in two key applications: the reconstruction of historical total factor productivity (TFP) and projections of potential output.

For our quantitative exploration, we feed forests ES data into the current tool underpinning the European macroeconomic surveillance framework, i.e. the European Unions' Commonly Agreed Methodology (EUCAM), that provides estimates of potential output over the medium term[8]. It relies on a production function-based approach, which features manufactured capital K_t and labour L_t as inputs as well as total factor productivity TFP_t as the Solow residual (Equation 4):

Equation 4. The Total Factor Productivity

$$Y_t = TFP_t L_t^{\alpha_1} K_t^{(1-\alpha_1)}$$

Following Dasgupta (2021), augmenting Equation 3 with natural capital, denoting with S_t the regulating and maintenance services provided by forest and with mN_t timber, yields:

Equation 5. The augmented Total Factor Productivity

$$Y_t = TFP_t S_t^{\beta} L_t^{\alpha_1} K_t^{\alpha_2} mN_t^{(1-\alpha_1-\alpha_2)}$$

where β defines the degree to which non-marketed natural capital underpins economic production.

When comparing the production function augmented with natural capital (Equation 5) to the standard Cobb-Douglas function used in the EUCAM framework (Havik et al., 2014), we observe that TFP is affected by the inclusion of two additional terms: marketed and non-marketed natural capital. Therefore, we can reconstruct historical TFP while accounting for natural capital.

Our results indicate that annual TFP growth changes significantly when natural capital dynamics are considered, with the magnitude of the effect depending on the calibration of production elasticities relative to its inputs. Additionally, our analysis reveals notable heterogeneity across EU countries. To assess the economic relevance of natural resources and their degradation in the medium-long-term we describe how simple macroeconomic tool might be used to provide potential output projections that take into account future dynamics of natural capital. Our reference is again the standard Cobb-Douglas production function (Equation 4), currently employed in EUCAM. As described above, this production function can be augmented along the lines of Dasgupta (2014) and Bastien-Olvera et al. (2023) to incorporate marketed and non-marketed natural capital, as presented in Equation 5. The

[7] WorldBank (2021). The Changing Wealth of Nations 2021: Managing Assets for the Future. Washington, DC: World Bank.

[8] Potential output and the output gap are key ingredients of the fiscal surveillance process which emanates from the Stability and Growth Pact, recently updated in the Economic Governance Review. The European Union's Commonly Agreed Methodology is discussed at the Potential Output Working Group of the Economic Policy Committee.

Cobb-Douglas production function assumes implicitly a high degree of substitution between natural capital and the other means of production. However, in the plausible case where ES are critical to production, this assumption should be relaxed. A standard way to tackle this issue is by considering a more general production function specification[9] which accounts explicitly for the elasticity of substitution (σ):

Equation 6. The augmented Total Factor Productivity with specifications on elasticity of substitution

$$Y_t = TFP_t \left(\omega S_t^{\frac{\sigma-1}{\sigma}} + (1-\omega)\left(L_t^{\alpha_1} K_t^{\alpha_2} m N K_t^{1-\alpha_1-\alpha_2}\right)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

where $\omega = \beta/(1+\beta)$. To simulate climate-change induced projections for marketed and non-marketed ecosystem benefits, one may follow Bastien-Olvera et al. (2023), who propose trajectories of country-level natural capital stocks under climate-change-driven global ES degradation scenarios[10]. Under the most favourable scenario, global warming is contained well below 2 degrees Celsius by the end of the century, whereas the most extreme one foresees an expected global mean temperature increase of 4.4 degrees (Lee et al. (2021)). The temperature change is then linked to losses of natural capital via the damage function estimated by Bastien-Olvera et al. (2023). Our preliminary figures show that potential output projections deteriorate dramatically in the least favourable scenario. Assuming that ecosystem services can only to a limited degree be replaced by fixed assets or labour, the negative impact of their degradation on economic output could, indeed, become very large. In view of the uncertainty about the actual elasticity of substitution, our numerical simulations should however be seen as illustrative, and our analysis as a proof of concept.

6.3 Gross Ecosystem Product in a general equilibrium model

Ecosystem services play a vital role in supporting economies, both directly and indirectly. However, traditional measures of economic growth, such as GDP, fall short in accounting for the economic benefits and environmental costs associated with these services (Dasgupta, 2021). To address this limitation, Ouyang et al. (2013) introduced the concept of Gross Ecosystem Product (GEP), which quantifies the monetary value of ecosystem services to the economy. In this section, we briefly discuss the new GEP module that is implemented as an extension to the global computable general equilibrium (CGE) model MAGNET. The latter is a modelling tool widely used by the European Commission to simulate the socioeconomic and environmental impacts of different policy initiatives.

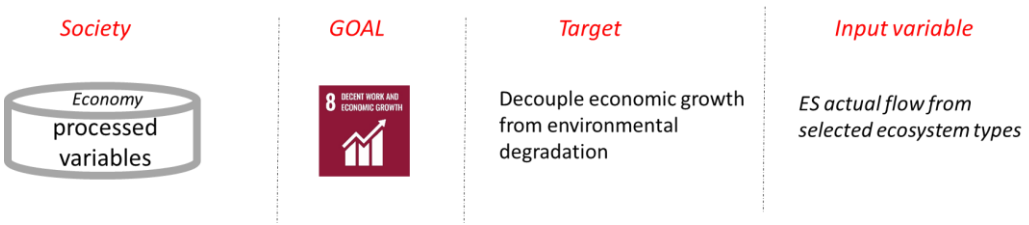
The GEP framework is aligned with the SEEA guidance. One of the advantages of the GEP is that it can be estimated alongside the GDP using national accounts data and allowing for a simultaneous

[9] The Cobb-Douglas production function discussed above is a special case of the Constant Elasticity of Substitution (CES) production function, which arises when the elasticity of substitution parameter σ equals 1. As σ tends to 0, the CES function approaches a Leontief form, where inputs are perfect complements, and as σ tends to infinity, it approaches a linear form, where inputs are perfect substitutes.

[10] Biome range shifts as projected by dynamic global vegetation models driven by general circulation climate models.

analysis of both indicators. The GEP indicator accounts for the economic value of ecosystem services, including those that are marketed, such as food and timber production. However, since these services are also included in the GDP, there is a duplication of efforts between the two metrics, as highlighted by recent studies (Polasky et al., 2023; Zhang et al., 2023). As in the previous case, the reference SDG is 8 related to economic growth, and in particular the target on decoupling economic growth from environmental degradation (Figure 31).

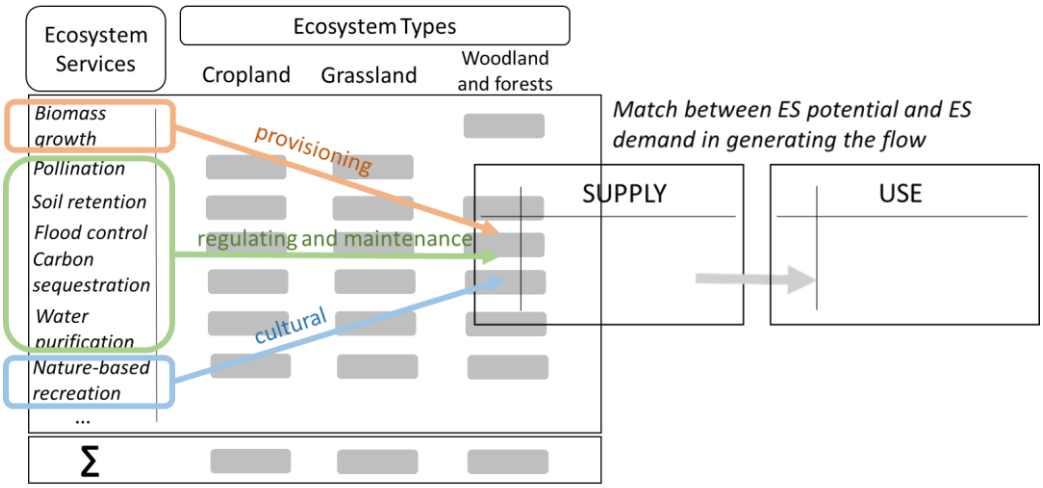
Figure 31. SDG: goal, target and input variable for introducing ES into the CGE model



Source: own elaboration

The Gross Ecosystem Product is calculated as the aggregate economic value of all ecosystem services provided within a country or region each year, encompassing three main components: Ecosystem Provisioning Services (EPV), Ecosystem Regulating and Maintenance Services (ERV), and Cultural Services (ECV). Hence, it can be expressed as GEP = EPV + ERV + ECV (Figure 32).

Figure 32. Input variable from Supply and Use Tables: the ES actual flow



Source: own elaboration

Following the above definition of GEP, all ecosystem services are simply additive. Consequently, the monetary value of GEP for different kinds of ecosystem services can be calculated as follows (Equation 7):

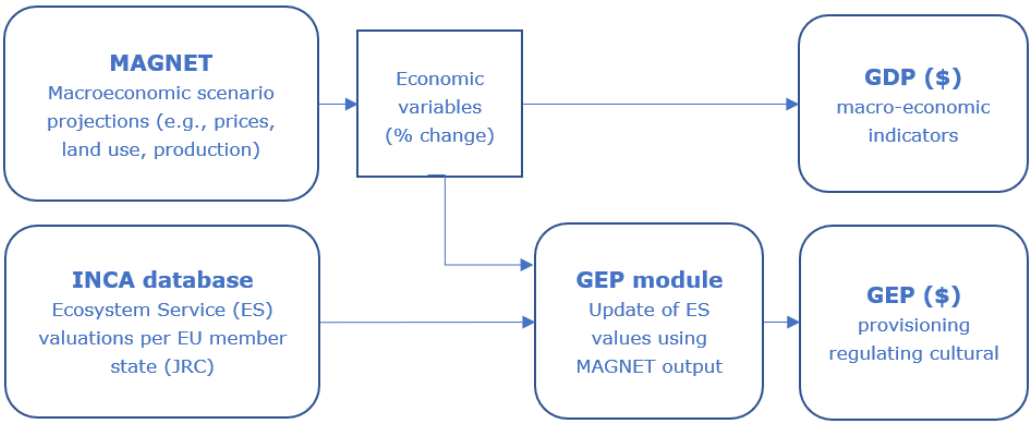
Equation 7. The Gross Ecosystem Product

$$GEP = \sum_i Y_i * Q_i * P_i + \sum_j Y_j * Q_j * P_j + \sum_k Y_k * Q_k * P_k$$

where i, j and k are sets of provisioning, regulating and maintenance, and cultural ecosystem services, respectively, Y is the proportion of accounting value attributable to nature, P is the accounting price of ecosystem service i, and Q is the quantity provided of ecosystem service i. For regulating and cultural ecosystem services, the entire value of the services is attributable to nature (Yi = 1). For provisioning there is a contribution from human labour and human-made inputs, so that Y < 1.

The GEP module is implemented as an extension to the global CGE model MAGNET - a recursive dynamic, multi-regional, multi-commodity CGE model, covering the entire global economy (Woltjer and Kuiper, 2014). Like other CGE models, MAGNET explicitly represents the economic linkages across the sectors of each regional economy. This is particularly important when analysing policy effects in sectors that are vertically linked with each other, such as bio economy sectors. The new GEP module utilizes the INCA dataset to establish the initial quantities and prices of ecosystem services (Vysna et al., 2021). Within the new module, GEP indicator is integrated and revised with economic change variables (such as land use and production) in MAGNET through a post-processing process. To assess the pressure on ecosystems we model the evolution of both the demand and the supply of ecosystem services.

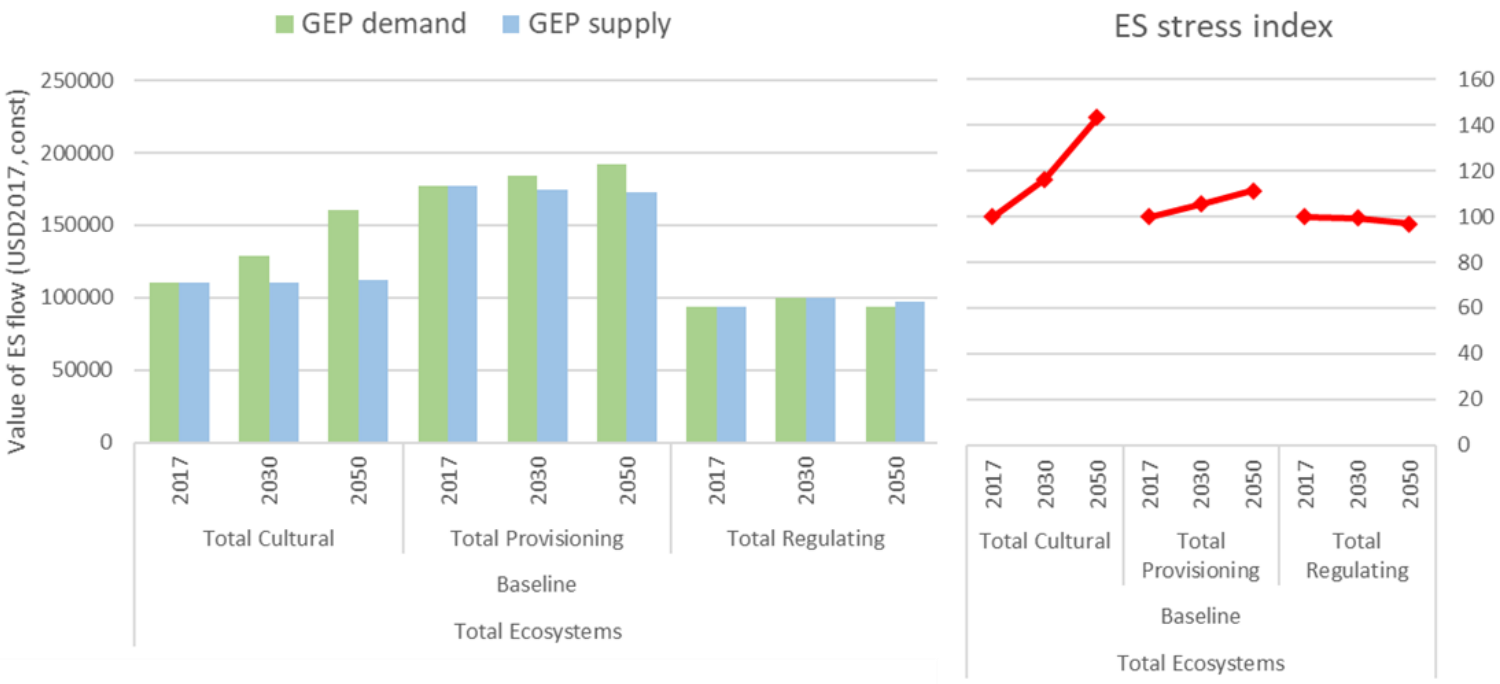
Figure 33. Procedure underpinning the GEP calculation



Source: Rokicki et al. (2024)

As an example of the application of the new module we present the results of the baseline projections until 2050 (based on SSP2, the middle of the road socioeconomic pathway). On the figure below, it can be observed that the demand and the supply of particular ecosystem services are growing at different rates (left panel). Consequently, there can be appreciated significant differences in the pressure on ecosystems. The latter is approximated by the ES stress index (right panel). It increases sharply for cultural services, rises slowly for provisioning services and slightly decreases for regulating services.

Figure 34. Results of the GEP application



Source: own elaboration

The above example shows that the new module can provide important insights on the evolution of GEP and the pressure on ecosystem services. The application of the module in simulations assessing different policies may, in turn, shed a new light on their environmental impacts from an economic point of view.

6.4 Natural capital and regional growth

The study available in Giannakis et al. (2025) emphasizes the significance of incorporating natural capital into regional development policies and offers evidence to inform future investments in biodiversity protection and ecosystem restoration, in line with the EU's green transition and cohesion objectives. A policy request has been submitted by DG REGIO to investigate the following aspects:

1. The dependency of regions on natural capital and the contribution of regional ecosystem services to the regional GDP.
2. The support of rural regions to urban or cities in terms of ecosystem services provision, considering interregional flows.

The objective of this request was to provide evidence for increasing the cohesion funds allocated to biodiversity investments during the next programming period (2028-2035). Investing in the protection and restoration of biodiversity will lead to sustainable provision of ecosystem services, which will be capitalised in the ecosystem assets through future flows, resulting in high natural capital values. If there is evidence of a strong connection between growth indicators and natural capital, biodiversity investment will yield high returns, justifying the allocation of cohesion funds to biodiversity for increasing regional growth and reducing regional disparities. Once again, the reference target, within SDG 8 on economic growth, is the one (8.4) on decoupling economic growth from environmental degradation (Figure 35).

Figure 35. SDG: goal, target and input variable for introducing ES into regional growth

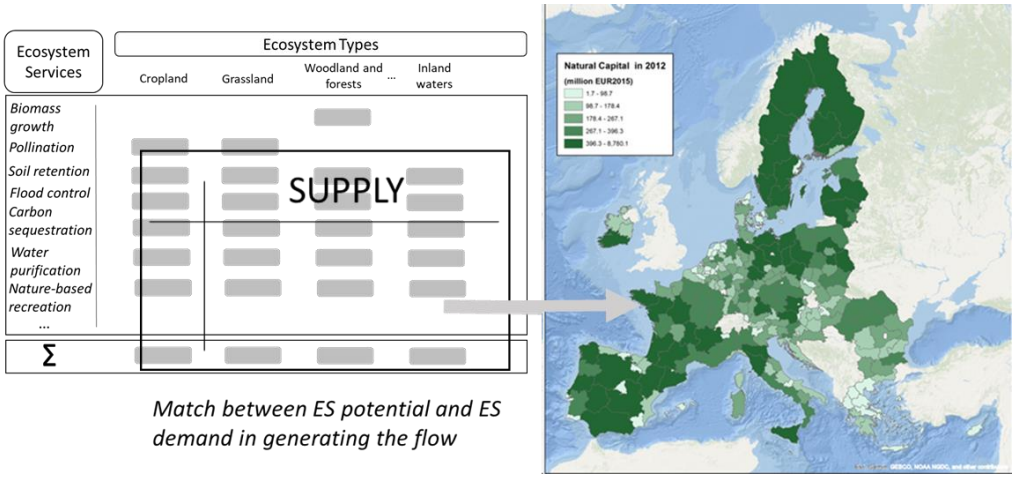
Source: own elaboration

To address the request from DG REGIO, it is necessary to assess the spatial dependence of natural capital distribution across regions. The connection between natural capital and economic growth has become a key focus, particularly in sustainable development and Europe's green transition strategy, as environmental degradation, biodiversity loss, and natural resource management concerns

intensify. Crucial to sustaining economic activity, natural capital encompasses resources and ecosystem assets, playing a vital role in regions directly reliant on agriculture, and forestry but also more broadly to other sectors that depend on nature indirectly. . Within the EU's sustainability agenda, ensuring economic development that doesn't harm natural ecosystems is a pressing need, particularly for the European Green Deal, EU Biodiversity Strategy for 2030, and Cohesion Policy revisions. The measurement and incorporation of natural capital into growth accounting frameworks is a central challenge in the literature, as traditional economic models often omit environmental factors. This leads to an incomplete understanding of regional economic performance and growth sustainability, misattributing the productive contribution of natural capital to capital, labour productivity, and Total Factor Productivity (TFP). Additionally, the spatial dependencies between regions are increasingly recognized as significant drivers in economic growth, but limited research has examined the spatial dependencies as well as the possible spill over effects of natural capital. Incorporating natural capital into growth models offers a more comprehensive perspective on regional economic development, recognizing the importance of ecosystems in delivering critical services and driving a shift toward sustainable economic practices within the EU's green transition goals. Despite its importance, the overall contribution of natural capital to regional economic growth remains underexplored.

The study represented natural capital services through ecosystem service flows. The empirical analysis covered seven ecosystem services namely (i) crop pollination, (ii) flood control, (iii) habitat and species maintenance, (iv) carbon sequestration, (v) soil retention, (vi) timber provision, and (vii) crop provision. Monetary estimates of these ecosystem service flows within EU NUTS 2 regions are sourced from the Integrated Natural Capital Accounting (INCA) project[11]. Consequently, the applicable accounting module is the supply and use tables in monetary terms, at the regional (i.e. NUTS 2) scale. Values are provided per ecosystem service, and they are aggregated by ecosystem type.

Figure 36. Input variable from Supply and Use Tables: extraction of zonal statistics of ES actual flow



Source: own elaboration based on Giannakis et al. (2025)

[11] https://ecosystem-accounts.jrc.ec.europa.eu/about-inca

To quantify the contribution of natural capital on regional economic growth, the study employed the Cobb-Douglas aggregate production function. The function was further specified to accommodate the potential endogeneity between natural capital and economic output and therefore lagged values of natural capital were used to mitigate these sources of endogeneity. Spatial dependencies in the distribution of our growth model's variables were measured using both a global Moran's I statistic and the local Moran's I test. The effect of spatial spill overs on regional growth was explored through the employment of a spatial regression model, i.e. a Spatial Durbin Model (SDM). The SDM, which accounts for global spill over effects through spatial lag terms of both the dependent variable and the explanatory variables, is formulated as:

Equation 8. Spatial Durbin model

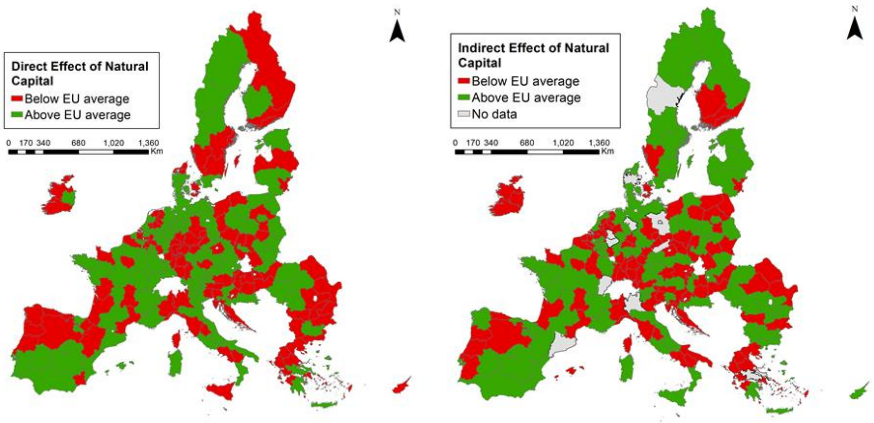
$$\ln y_{it} = a_0 + a \ln l_{it} + b \ln n_{it-1} + c \ln k_{it} + \rho W y_{it} + W ln\Omega_{it}\theta + a_i + \delta t + e_{it}$$

where:
y_{it} represents the gross value added (GVA) in region i in accounting period t $(i = 1, \dots, 222; t = 2006, 2012, 2018; t - 1 = 2000, 2006, 2012)$;
l_{it} represents the compensation of employment in region i in year t ;
$n_{(it-1)}$ represents the value of ecosystem services in region i in year $t - 1$;
k_{it} represents the capital stock in region i in year t ;
a , b , and c are coefficients assessing the effects of labour, natural capital, and produced capital, respectively, on regional economic growth;
ε_{it} is the error term; all units are defined in constant 2010 prices (euro);
ρ denotes the spatial autoregressive parameter, indicating the effect of GVA in neighbouring regions on a region's own GVA;
W is a row-standardised spatial weights matrix used to model spillover effects across regions;
ϑ represents the spatial parameters of the spatially lagged variables Ω_{it};
a_i captures the region-specific fixed effects;
t represents the time trend;
δ captures the average change in GVA over time.

The results of the SDM for the augmented Cobb-Douglas production function reveal the positive impact of natural capital on regional economic growth. A 10% increase in natural capital value results in a 0.4% increase in EU-27 NUTS 2 regions' growth. These findings confirm the statistically significant positive contribution of natural capital to regional economic growth and indicate that omitting natural capital from growth models can lead to overestimating the contributions of labour and capital. The significant spatial lag coefficient of natural capital demonstrates that neighbouring regions with high natural capital positively influence own-region economic growth, showcasing positive spill over effects. A 10% increase in the natural capital value of adjacent regions results in a 0.3% increase in EU-27 NUTS 2 regions' GVA. The total effect of natural capital on regional economic growth, considering both direct and indirect effects, is approximately 0.07. This implies that a 10% increase in EU-27 NUTS 2 regions' natural capital value leads to a 0.7% increase in their GVA during the study period. In absolute terms this implies that an increase of 9.3 billion Euros in natural capital value, equivalent to a 10% average annual value and comparable to the 2014-2020 cohesion policy's biodiversity investment (11 billion Euros), would enhance EU-NUTS 2 regions' GVA by 71.4 billion Euros. It is vital to note that these estimates may underestimate

natural capital's contribution to regional economic growth, as key services such as cultural services are not factored into the natural capital valuation due to a lack of regional-level data.

Figure 37. Natural capital's direct and indirect (spill over) effects across EU NUTS 2 regions



Source: Giannakis et al. (2025)

To deepen the understanding of natural capital's contribution, interaction terms were introduced in the SDM model, allowing for a more detailed examination of its direct and spill over effects on regional economic output across different regions. Figure 37 highlights EU-NUTS 2 regions with above-average direct and spill over effects on economic growth, indicating where investments in natural capital yield higher returns. Targeting these regions fosters regional cohesion and amplifies economic impact across a broader geographic area. Conversely, regions with below-average effects may require more targeted interventions to unlock their full potential. Based on the obtained outcomes, this kind of indicator can eventually also be linked to and support SDG target 15.a "Mobilize and significantly increase financial resources from all sources to conserve and sustainable use biodiversity and ecosystems.

7 Conclusions: paving the way forward

Ecosystem services accounts can be used in many ways by different users for different purposes. They can be used as conventional statistics to analyse values at a given moment or to measure trends over time. They can be used for processing and compiling indicators or as input variables in economic and financial tools. The applications described in this report show that compiling ES accounts is only one way of usage. Indeed, given the complexity of bringing together ecological and accounting domains and bridging ecology and economy, the role of compilers may go beyond compilation to support interpretation and implementation.

Acknowledging the value of ecosystem services implemented in socio-economic analyses, policies and decision-making is crucial to implement the 2030 Agenda for Sustainable Development. By understanding and promoting the linkages between ecosystem services and Sustainable Development Goals (SDGs), we can foster a mutually beneficial measuring system that connects people and nature.

The potential uses of ES accounts are much broader than presented in this report. To explore the possibilities, some barriers and limitations need to be overcome. A common problem is that avenues need to be paved to allow connecting INCA data to conventional economic models (see Hirschbuehl et al., 2025 for detailing a way forward on introducing INCA ES into Input Output models based on Encore sectoral exposures). All applications in this report are based on the INCA accounts, which are characterised by the ES match and mismatch approach. This in turn allows for additional applications that otherwise would not be possible.

First, there are several limitations in the methodology underlying the biophysical assessment of ES within INCA. Current models and their underlying data sources have been designed for the European context. Many environmental issues have a perspective that extends beyond Europe, so other tools and ES assessments may be preferable in such cases. This can be a matter of both modelling scope and data availability. In addition, all current models do not include the calculation of uncertainty, e.g. data lacks error bars. The more ES accounts are used, the more uncertainty analysis becomes increasingly important.

Second, an intrinsic problem of ES accounts lies in the way ES are assessed. Despite the standardization of the classification and of the accounting format, there is no standardization of the assessment criteria or of the operational procedure to be followed. This creates difficulties when it comes to comparing, aggregating and interpreting changes in ES accounts, which are assessed using different approaches and methodologies.

Third, an additional barrier common to all ES accounts is their lack of usability for non-accountants. The lack of usability concerns both the content of the accounts, which is not intuitive to capture, and the format of the information (spatial and tabular) which is not user friendly. Additional intermediate steps should be taken to enable users (who are not experts in natural capital accounting) to get the most out of the accounts. In addition, as they are a way of collating data rather than a final product, they can be used by non accountants and can be complimentary to other frameworks (Xiao et al., 2024). Most importantly, the underlying data should be made available to decision-makers, as these may be particularly useful across a range of purposes, including ecosystem maps, and spatial data on ecosystem condition and supply and use of ecosystem service.

Once the limitations and barriers have been addressed, the use of ES accounts in economic and financial research topics can begin in a more systematic way. This will facilitate the development of

a range of proposals to address the research agenda. These include the calculation of an ecologically adjusted net value added, which considers the input-output supply chain, or the Gross Ecosystem Product indicator, complementing the GDP in macroeconomic modelling. Another proposal is nature-related financial disclosures for companies and the ES accounts in monetary terms as the basis for the calculation of biodiversity credits. Studying the bio-physical underpinnings of ecosystem dynamics (e.g. use, depletion, and natural regeneration) would allow basing scenarios on macroeconomic impact of ecosystem degradation on explicit accumulation equations for natural capital. Ideally, by also including the impact of pollutants on ecosystem dynamics, the feedback loop between economic activities, damages to ecosystems and consequences for economic prospects could be closed.

Finally, to promote a systematic use of ES accounts by both the public and private sectors, we need to prepare a global ecosystem services database. One way to proceed is to compare the INCA numbers with INVEST numbers[12], which has a global geographical coverage. Then, there is the need for systematic updates of the data based on exactly the same methodology. This would allow both to evaluate the evolution of ecosystems and apply the data to economic studies that require panel data.

[12] Ref. https://naturalcapitalproject.stanford.edu/

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List of abbreviations and definitions

Abbreviations	Definitions
CGE	Computable general Equilibrium Model
CICES	Common International Classification of Ecosystem Services
FAO	Food and Agriculture Organization
EC	European Commission
ECB	European Central Bank
EU-NUTS	European Union - Nomenclature of Territorial Units
ENCORE	Exploring Natural Capital Opportunities, Risks and Exposure
ES	Ecosystem Services
ESRB	European Systemic Risk Board
GFCM	General Fsheries Commission for the Mediterranean Area
GSA	Geographical Sub-Area
GVA	Goss Value Added
INCA	Integrated system for Natural Capital Accounts
JRC	Joint Research Centre
NPV	Net Present Value
SDGs	Sustainable Development Goals
SDM	Spatial Durbn Model
SEEA CF	System of Environmental and Economic Accounting – Central Framework
SEEA EA	System of Environmental and Economic Accounting – Ecosystem Accounts
SME	Small and Medium Enterprises
SUT	Supply and Use Tables
TB_SGd	Trophic based Segrass dependency
TL	Trophic level
TFP	Total Factor Productivity
UNCEEA	United Natiorn Committee of Experts on Environmental and Economic Accounts
UNSD	United Nations Statistical Division
UN	United Nations

List of figures

Figure 1. Stylized visualization on how ES accounts can directly support some policy actors, such as statistical offices, policy actors that directly process indicators and international organizations that collect data; or indirectly support other policy actors through the work of other experts working in specific disciplinary areas such as macroeconomics, central banking, agricultural and fisheries policies 6

Figure 2. Ecosystem Service (ES) actual flow generated by the interaction between the potential supplied by the ecosystem and the demand from the socio-economic system. 9

Figure 3. Ecosystem Service (ES) vulnerability generated by a demand from the socio-economic system that cannot be covered by the potential supplied by the ecosystem. 10

Figure 4. Tables that form the overall accounting framework: red signs indicate where to account for factors that may negatively affect the production of ecosystem services and determine a shortfall in provision. 11

Figure 5. Simplified representation of natural resource asset accounts with a focus on the "addition" item, according to the SEEA CF 12

Figure 6. Simplified representation of ecosystem asset accounts in monetary terms, ... 13

Figure 7. Simplified diagram of the convergence model Figure title is inserted above the figure. .. 14

Figure 8. The SDG Wedding Cake 16

Figure 9. ES accounts to provide descriptive statistics to measure and monitor the biosphere..... 17

Figure 10. ES accounts to process indicators that measure the interaction with society 19

Figure 11. ES accounts to provide input variables into economic and financial models, tools, frameworks 20

Figure 12. Connection between the uses of Ecosystem Services accounts and the Sustainable Development Goals: some examples..... 22

Figure 13. SDG: goal, broad description and kind of analysis for the cases of basic descriptive analysis..... 23

Figure 14. SDG: goal, broad description and data for the cases of basic descriptive analysis 23

Figure 15. ES supply in % of the aggregated value in monetary terms (2018) 24

Figure 16. % change in ES actual flow between the reported accounting periods (*) in physical terms..... 25

Figure 17. The B.1 index values for EU data: A) overall Index (all ES types, in black, n=8), Provisioning (red, n=2), Regulating and Maintenance (blue, n=5), and Cultural services (green, n=1).; B) Trends for ES types are coloured by Provisioning (red-orange), Regulating and Maintenance (blues), and Cultural (greens). Note scale of Y axis differs between index and ES specific trend plots 27

Figure 18. SDG: goal, target and novel indicator for the seagrass dependency of fish provision..... 29

Figure 19. Processed information from Supply and Use Tables: ES match 30

Figure 20. Trophic-based seagrass dependency (TB_SGd) 31

Figure 21. Direct and indirect use of seagrass for fish provision and the complementary value that could be at risk if seagrass (as primary trophic level) were not there (year 2018, EU member states bordering the Mediterranean Sea for 5 selected fish species) 32

Figure 22. SDG: goal, target and novel indicator for ES Vulnerability Index for the agricultural sector 33

Figure 23. Processed information from Supply and Use Tables: ES mismatch 33

Figure 24. Components of the ES Vulnerability Index in the EU Member States with the highest Gross Value Added and Exports in agricultural products 34

Figure 25. SDG: goal, target and input variable for SME's exposure to ES 35

Figure 26. Input variable from Supply and Use Tables: ES mismatch 36

Figure 27. Scheme of the basic method 37

Figure 28. Geospatially refined risk exposure 38

Figure 29. SDG: goal, target and input variable for introducing ES into the macroeconomic framework 39

Figure 30. Input variable from Supply and Use Tables: the Net Present Value of ES actual flow 39

Figure 31. SDG: goal, target and input variable for introducing ES into the CGE model 42

Figure 32. Input variable from Supply and Use Tables: the ES actual flow 42

Figure 33. Procedure underpinning the GEP calculation 43

Figure 34. Results of the GEP application 45

Figure 35. SDG: goal, target and input variable for introducing ES into regional growth 46

Figure 36. Input variable from Supply and Use Tables: extraction of zonal statistics of ES actual flow 47

Figure 37. Natural capital's direct and indirect (spillover) effects across EU NUTS 2 regions 49

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